



ORIGINAL RESEARCH ARTICLE

Optimizing Fused Deposition Modeling Process Parameters to Enhance Dimensional Accuracy and Tensile Strength of Bio-based Polylactic Acid in Additive Manufacturing

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Achieving both high dimensional accuracy and mechanical strength in fused deposition modeling (FDM) remains a significant challenge, particularly for sustainable materials like bio-based polylactic acid (BB-PLA) which exhibits distinct thermal and mechanical behavior compared to conventional PLA. This study presents a dual-objective optimization of seven key FDM process parameters—infll percentage, layer thickness, nozzle diameter, material temperature, raster orientation, build orientation, and extruding speed—to simultaneously enhance geometric fidelity and tensile strength. A Taguchi L18 orthogonal array was employed to design experiments efficiently, and performance was evaluated using dimensional deviation metrics and standardized tensile testing. Analysis of variance identified build orientation, nozzle diameter, and infll percentage as the most influential parameters affecting both output objectives. The optimal combination—on-edge orientation, 0.5 mm nozzle diameter, and 80% infll—yielded a maximum tensile strength of 59.84 MPa and a minimal dimensional error of 0.16%. These findings were validated through confirmation testing and regression modeling ($R^2 = 0.88$), which showed high predictive accuracy. These performance enhancements were further validated through microstructural analysis, which revealed enhanced crystallinity, reduced porosity, and superior interlayer adhesion in specimens produced under optimized conditions. The results show that BB-PLA can be converted from prototyping material to a functional candidate material that can be used for precision applications as a result of strategic optimization of the process. From this research, a validated design framework to further sustainable advancements in additive manufacturing is presented that concurrently tackles discrepancies in dimensional accuracy and structural outcomes.

Keywords additive manufacturing, bio-based polylactic acid, dimensional accuracy, effective process parameter, fused deposition modeling, optimization of parameters, Taguchi method, tensile strength

1. Introduction

Additive manufacturing, particularly fused deposition modeling, has become widely adopted for its low cost, design flexibility, and ability to fabricate complex parts without tooling (Ref 1). Fused deposition modeling (FDM) is widely applied in aerospace, biomedical, automotive, and customer-delivered goods industries in which rapid prototyping and customization are applications (Ref 2, 3). However, the disappointing point of FDM is the geometric accuracy. Often, the printed parts will have differing properties from those models produced in the

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List of Symbols

ρ	Material density
δ_d	Thermal deformation-induced error
δ_t	Thermal diffusion depth
ε_T	Thermal expansion-induced dimensional error
ϕ	Infill density
α_T	Thermal expansion coefficient
ΔT	Temperature difference between the extrudate and ambient
n_g	Number of infll grid cells
ΔL	Dimensional deviation
$\sigma_{\parallel}$	Tensile strength along filament direction
$\sigma_{\perp}$	Tensile strength across filament direction
θ	Raster angle
$\varepsilon_{\parallel}$	Shrinkage along filament direction
$\varepsilon_{\perp}$	Shrinkage across filament direction
σ_{eff}	Effective tensile strength
k_r	Resolution coefficient
ε_v	Dimensional error
β_1	Material specific constant
β_2	Printer specific constant
λ	Extrusion expansion factor
A_e	Extruded bead cross-sectional area
A_f	Filled area per cell
A_t	Total cross-sectional area

c_p	Specific heat capacity
F_α	F-value at 99% confidence
k	Thermal conductivity
k_r	Resolution coefficient
$L_{\min}$	Minimum printable feature size
n_l	Number of layers
T_a	Ambient temperature
t_c	Thermal contact time
V_e	Error variance
Y_{opt}	Optimum settings
A	Infill percentage
B	Layer thickness
C	Material temperature
D	Build orientation
E	Raster orientation
F	Nozzle diameter
G	Extruding speed
H	Component height
ABS	Acrylonitrile Butadiene Styrene
ANOVA	Analysis of Variance
ASTM	American Society for Testing and Materials
BB-PLA	Bio-Based Polylactic Acid
CAD	Computer-Aided Design
CI	Confidence Interval
DOF	Degrees Of Freedom
PLA	Polylactic Acid
SEM	Scanning Electron Microscopy
S/N	Signal-to-Noise
T/N	Grand mean of the S/N ratios
TEM	Transmission Electron Microscopy
XRD	X-ray Diffraction

CAD software, caused by thermal contraction or related effects such as anisotropy and inconsistencies in the depositions (Ref 4). In high-precision applications, where structural integrity, functional performance, or fit for purpose is crucial (e.g., medical implants or functional assembly), any dimensional inaccuracies may lead to structural instability, malfunction, or improper functionality (Ref 5, 6). At the same time, mechanical behavior (especially tensile strength) is even more important, which depends not only on the base material properties but also on interlayer bonding, void formation, and matrix filament orientation as a function of the printing process (Ref 7, 8).

These difficulties are increased in bio-based PLA, which is different from conventional PLA in crystallization, shrinkage behavior, and interlayer adhesion (Ref 9-11). While BB-PLA is interesting because of its sustainable aspect, it frequently experiences reduced mechanical strength in comparison to polynomials based on petroleum, such as Acrylonitrile Butadiene Styrene (ABS) or polyethylene terephthalate glycol (Ref 12-14). Therefore, there is an urgent need to find process conditions that combine simultaneously dimensional accuracy and high tensile strength so that BB-PLA could be developed from prototyping material to a possible solution for functional, load-bearing applications. This study helps in closing this gap by systematically studying the role of the FDM parameters and developing a dual objective optimization model tailored to BB-PLA.

Several investigations have explored the processing parameter dependencies of dimensional fidelity, tensile strength, and surface finish in addition to several other properties of FDM printed parts, but most of them have focused on petrochemical-based polymers such as ABS and conventional PLA (Ref 15, 16). For example, in Ref 17, the material temperature, printing speed, and layer depth were used as independent control variables to assess the dimensional accuracy of ABS parts, and solid conclusions about parameter interaction were reached. However, ABS has a different thermal expansion and shrinkage behavior than PLA, and such results cannot directly be translated to bio-based counterparts such as BB-PLA. Similarly, Ref 18 optimized cost-efficiency and dimensional accuracy using an L9 orthogonal design, but again the focus on petrochemical polymers means a lack of insight about how analogous strategies apply to renewable, biodegradable polymers with different thermal cycling properties. Some efforts have been made to get beyond ABS by examining modified or composite PLA systems. For example, Ref 19 presented an analysis of PLA reinforced with nanographene, proving that composition, erasure angle, and infill density have a direct impact on wear resistance. However, this study dealt mostly with the filler reinforcement than the intrinsic behavior of pure PLA or BB-PLA. Other attempts, like in Ref 20 were focused on laser polishing of acrylonitrile styrene acrylate to minimize roughness and enhance strength and the need for post-processing, but did not consider parameter optimization at the printing stage. Ulkir and Ersoy (Ref 21) presented a detailed view of the process-structure-property relationships in FDM printed PLA composites and showed how microstructural features like layer bonding and crystallinity have a direct impact on mechanical performance and justify the necessity of an integrated structural analysis study for optimization. Likewise, Ref 22 reported the effect of temperature on warpage in petrochemical polymers, while Ref 23 investigated stainless steel-PLA composites, including the melt flow and clogging of the nozzle problems that are unique to the metal-polymer hybrids. These studies help to characterize the extensive FDM studies carried out; however, they also reveal how their results are highly material specific, and thus not directly applicable to BB-PLA.

Investigations of PLA more closely related to the scope in this work are Ref 24-26, which analyzed PLA using different experimental designs such as response surface methodology, Box-Behnken design, and factorial. They continued to prove that parameters such as layer thickness, infill density, and raster angle have a strong effect on tensile strength and dimensional results. For example, the effects of reduced layer thickness and reduced raster angle on tensile strength and surface finish were verified, and efficient statistical techniques such as ANOVA and regression were used to give predictive insight into parameter-property relationships. While these studies underline the need for optimization for PLA, they consider the optimization of dimensional accuracy and mechanical strength mainly as independent targets, optimizing each target separately (not combining the two). This need is further supported by recent work by Yorke and Chowdary (Ref 27), who showed that systematic tuning of parameters in FDM processes can indeed be used to attain a multi-objective optimization of PLA with simultaneous improvement in both mechanical strength and dimensional accuracy. A similar treatment of the optimization of FDM processes for PLA/wood biocomposites was also shown by Morvayova et al. (Ref 28) using a multi-objective

approach due to complex relationships between natural fillers, process parameters, and part quality. Their hybrid methodology, including Taguchi orthogonal array and Gray relational analysis, could efficiently eliminate voids and enhance the dimensional accuracy, which pointed out the shortcoming of the single-objective optimization methods in finding reliable solutions of biocomposite materials. In this respect, a recent publication by Kaplan et al. (Ref 29) showed the importance of sophisticated in-line monitoring and machine-learning methodologies for predicting the mechanical behavior of FDM printed parts. Their BiLSTM-Bayesian optimization scheme obtained excellent accuracy in predicting Shore D hardness and surface roughness of thermoplastic polyurethane-based orthoses and showed layer thickness as the most contributing variable. This shows the importance of the parameter-induced variability as an essential aspect needing robust process design rules for polymer additive manufacturing.

In summary, the literature shows that while ABS and conventional PLA have been extensively researched, the research on BB-PLA is still underrepresented, although it has become an important green alternative in today's market. Due to its bio-modified nature, the thermal behavior, the crystallinity patterns, and interlayer bonding behavior of BB-PLA are often observed to be different from normal PLA. These differences may imply that optimization strategies developed for ABS or PLA cannot be expected to apply directly for BB-PLA. Moreover, although earlier works have normally maximized dimensional accuracy or tensile strength, respectively, there exists a gap regarding the comprehensive dual-objective optimization for biodegradable polymers. As a result, the current study systematically explores the coupling effect between FDM parameter and dimension fidelity, along with mechanical performance of BB-PLA components. While these properties have been optimized individually for conventional polymers in previous investigations, in this work we provide a thoroughly developed dual-objective optimization framework for BB-PLA in which mechanical testing, dimensional analysis, and advanced microstructural characterization are combined to clarify the underlying mechanisms.

The paper is presented in the following way: section 2 describes the FDM parameters taken into account and their relationship with dimensional accuracy and mechanical strength. The experimental design and procedures are explained in section 3. Section 4 provides the results of dimensional accuracy and mechanical properties analysis. Section 5 draws conclusions from the major findings and proposes specific directions for future work.

2. Key Process Parameters in FDM

In FDM, several process parameters directly influence the tensile strength of the printed parts by affecting interlayer adhesion, material consolidation, and load distribution. This study examines the effect of the following parameters: infill percentage, layer thickness, diameter of the nozzle, material temperature, raster orientation, build orientation, and extruding speed.

2.1 Infill Percentage (A)

Infill percentage is the percentage of volume occupied by solid material, or the percentage of internal volume of the printed component, which simultaneously affects the accuracy of dimensions and the mechanical strength of the printed component.

From the dimensional accuracy angle, more infill seems to change local thermal fields when depositing, which will affect shrinkage rates and warping, and particularly near solid boundaries. These effects can result in dimensional variabilities from nominal dimensions on parts having thin walls or high aspect ratios. A simplified equation to represent the dependence of thermal deformation-induced error δ_d on infill density ϕ can be represented as

$$\delta_d \propto \alpha_T \cdot \phi \cdot \Delta T \quad (\text{Eq 1})$$

where α_T is the thermal expansion coefficient of the material, ϕ is the density of infill (a unitless fraction from 0 to 1), and ΔT is the difference in temperature between the extrudate and the surroundings. The physical reason behind this expression is that as the infill density increases the bulk thermal mass increases, so that cooling contraction effects increase. Additionally, if cooling is not ideally controlled then warping and dimensional deviation can increase with increased infill.

From a mechanical point of view, higher infill percentages deliver higher effective material density (ρ_{eff}) which results in higher load bearing capacity of the component and lower internal voids. Such greater intra-molecular cohesion makes it more uniformly stressed and makes it more resistant to cracking under tensile (pulling) stress. The effective density of the printed object can be modeled as (Ref 30):

$$\rho_{\text{eff}} = \rho_{\text{material}} \times \left(\frac{n_g \cdot A_f}{A_t} \right) \quad (\text{Eq 2})$$

where ρ_{material} is the density of BB—PLA material, n_g are the bed infill grid cells, A_f is the cell area, and A_t is the total cross-sectional contact. Physically this relation indicates that as more of the internal grid is filled in the density of the part gets closer to that of solid BB-PLA to improve the distribution of stresses and the tensile strength. When $\frac{A_f}{A_t} \rightarrow 1$, $\rho_{\text{eff}} \rightarrow \rho_{\text{material}}$, which indicates max-infill (100%). However, when the material increases beyond the optimum infill value, the concentration of the material can cause heat to build up and internal stresses to be generated due to poor thermal dissipation, which will infringe on the integrity and dimensional stability of the part.

Thus, infill percentage must be optimized not just to increase the strengths of the part structurally but also to maintain geometrical accuracy. This balance is of special concern in sustainable polymers such as BB-PLA, where thermal behavior plays a large role in determining the strength and accuracy.

2.2 Layer Thickness (B)

Layer thickness is a critical factor that influences both the dimensional accuracy and mechanical performance of FDM-printed components, particularly along the Z-axis. In terms of strength, thinner layers narrow the contact between adjacent depositions, which has improved interlayer fusion, i.e., enhancement of tensile strength in the vertical direction. But if the layers are too thin, they can be impossible to bond, and the resulting structure can be weakened.

From a dimensional accuracy point of view, layer thickness defines the level of accuracy with which a 3D model is transferred to the physical part. During slicing, the CAD model is sliced into layers, and if the difference between the CAD model height and the thickness of the selected layer slices is not zero, then there will be rounding errors, which will cause dimensional deviations in the Z direction. The number of layers n_1 in a part of height H is given by

$$n_1 = \frac{H}{\text{layer thickness}} \quad (\text{Eq 3})$$

This simple geometric relation reflects how finer layers increase the resolution of the Z axis representation. The dimensional error in the Z axis of the layer thickness can be modeled as:

$$\varepsilon_z = H - n_1 \cdot B \quad (\text{Eq 4})$$

where B is the layer thickness and H is the desired height of the part. Physically, this equation arises because each mismatch between the intended and printed layer contributes to a stack-up error. Obviously, when the part height is not evenly divisible by the layer thickness, the printer then adjusts such that the last layer is either truncated or extended, which, of course, adds nonuniformity.

Hence, the selection of layer thickness is a compromise between two diametrically opposed requirements: minimization of the cumulative error to become dimensionally correct and maximizing the strength of interlayer bonds to become mechanically strong. Careful adjustment of this parameter is deemed necessary for the yield of high quality BB-PLA parts.

2.3 Material Temperature (C)

Material extrusion temperature is an important FDM material printing parameter, and it has a direct impact on interlayer bonding and dimensional precision. The extrusion temperature needs to be high enough to create sufficient diffusion of polymer chains between layers of polymer but not high enough to cause the polymer to degrade or to have excessive thermal setting.

From thermal expansion perspective, higher extrusion temperature increases the dilemma of warpage and excess dimensions focusing on the build plane. The thermal expansion-induced dimensional error ε_T can be modeled as

$$\varepsilon_T = \alpha_T \cdot \Delta T \cdot L \quad (\text{Eq 5})$$

where α_T is the thermal expansion coefficient of BB-PLA ($1/^\circ\text{C}$), $\Delta T = T_{\text{melt}} - T_{\text{ambient}}$. T_{melt} is the melting temperature, T_{ambient} is the ambient temperature and L is the initial dimension (mm). Higher values of ΔT increase the likelihood of post-deposition shrinkage, especially if cooling is uneven. This situation can lead to dimensional variations, edge curling or internal stresses which would affect geometric fidelity. The physical mechanism causing this phenomenon is due to the amplification of expansion in the deposition process and the shrinkage in the cooling process if the thermal gradients are strong in nature resulting in the thermal gradient warping, dimension overshoot, and residual stress.

Enhanced interlayer fusion is achieved by enhanced thermal overlap and associated effective bonding between extruding adjacent filaments. The thermal diffusion depth δ_t that deter-

mines the degree of heat penetration into the underlying layer during deposition is given by Ref 31

$$\delta_t = \sqrt{\frac{k}{\rho c_p G}} \quad (\text{Eq 6})$$

where k is the thermal conductivity of BB-PLA ($\text{W/m} \cdot \text{K}$), ρ is the material density (kg/m^3), where c_p represents the specific heat capacity ($\text{J/kg} \cdot \text{K}$), and G refers to the extrusion speed (mm/s). This expression comes directly from classical heat conduction theory and shows that temperature, material properties, and rate of deposition all combine to determine the magnitude of heat transfer. A greater depth of diffusion (δ_t) means greater thermal overlapping between sections of adjacent filaments to induce polymer-carbon monoxide chain entanglement and to increase layer adhesion. Conversely, excessive temperature above the degradation point of PLA (in general $> 210^\circ\text{C}$) causes chain scission and porosity and, therefore, part integrity is lost. Consequently, an optimal extruder temperature must fall within a thermal processing window in which interlayer diffusion is adequate to provide strong mechanical bonding and low levels of dimensional distortion occur during this process; this thermal processing window is especially narrow for semi-crystalline biopolymers such as BB-PLA, requiring careful control of thermal treatment to produce high-quality prints.

2.4 Build Orientation (D)

Build orientation defines the direction in which layers are deposited relative to the part's geometry and loading direction. It has a profound effect on both tensile strength and dimensional accuracy. Additionally, it impacts vertical and horizontal shrinkage behavior. In the case of fused deposition modeling, layer-wise deposition and the consequent disparity in the rate of cooling of the print from the bottom to the top are the causes of shrinkage anisotropy. The dimensional deviation on the build axis can be modeled as

$$\Delta L = \alpha_T \cdot \Delta T \cdot L_{\text{build}} \quad (\text{Eq 7})$$

where ΔL represents the dimensional deviation, α_T refers to the coefficient of thermal expansion, ΔT represents the temperature changes that occur during cooling and L_{build} represents the height in the build direction. On a physical level, this relation comes directly from thermal expansion theory, where taller builds in the Z-direction accumulate greater deviations. As an example, flat orientations (X) tend to result in greater vertical warping versus vertical warping of the on-edge orientations (Y), which tend to be distributed in far more varied, e.g., the width or depth.

From a mechanical perspective, build orientation controls the loads (given) in relation to deposited filaments. When printed in the on-edge orientation (Y-axis), the filaments lie along the loading axis to enhance interlayer bonding and suppress the stress concentrations. By contrast, in the flat orientation (X-axis), the loading will be through the layers where adhesion is weaker, often resulting in delamination of the interface. The mechanical response can be represented using an orientation-dependent strength model.

$$\sigma_{\text{eff}} = \sigma_{\parallel} \cdot \cos^2 \theta + \sigma_{\perp} \cdot \sin^2 \theta \quad (\text{Eq 8})$$

where $\sigma_{\parallel}$ is the tensile strength parallel to the filament, $\sigma_{\perp}$ is the tensile strength perpendicular to the layer(s), and θ is the angle of in-plane construction direction and loading direction.

For on-edge orientation, $\theta \approx 0^\circ$, maximizing $\sigma_{\text{eff}} \approx \sigma_{\parallel}$.

For flat orientation, $\theta \approx 90^\circ$, thus $\sigma_{\text{eff}} \approx \sigma_{\perp}$, which is typically lower.

Orientation also affects the way dimensional deviations are concentrated; as in flat orientation, the deviations are concentrated along Z, prone to vertical warping, whereas in on-edge builds, there may be warpage across the width or the depth of the part, changing the tolerance in different ways.

2.5 Raster Orientation (E)

Raster orientation refers to the angle at which the extruded filaments are laid down within each printed layer relative to the part's principal axis. This has a major control effect over the geometric accuracy and crucially over the anisotropy of the tensile strength, when working on elements undergoing a certain orientation of the load and, especially, when fine detail reproduction takes place.

From a geometrical point of view, raster orientation can cause directional warping or deviation of the edges, mainly close to sharp corners or unsupported overhangs. Different raster angles thereby also lead to different local heat distribution and shrinkage behavior, and hence have an effect on the dimensional stability. An estimate of anisotropic shrinkage-caused error ε_{θ} depends on the raster angle as given as follows:

$$\varepsilon_{\theta} = \varepsilon_{\parallel} \cos^2 \theta + \varepsilon_{\perp} \sin^2 \theta \quad (\text{Eq 9})$$

where $\varepsilon_{\parallel}$ is the shrinkage normal to the direction of the filament, $\varepsilon_{\perp}$ is the shrinkage across the direction of the filaments. Usually $\varepsilon_{\perp} > \varepsilon_{\parallel}$ because of the lowering of the bonding and the higher tendency of cooling contraction of road path. The physical basis of this relation is based on orthotropic shrinkage behavior; cooling is more uniform and restrained along filament paths and weaker bonding along filaments across paths and higher contraction will cause larger deviations. As a result, the dimensional accuracy depends on the re-distribution of these anisotropic effects, depending on the orientation of the raster.

When rasters are parallel to the loading direction (e.g., 0° raster angle), the printed filaments directly resist the tensile forces and create a higher ultimate tensile strength. Conversely, when rasters are oriented at oblique or perpendicular angles (e.g., 45° or 90°), load transfer occurs across inter-filament bonds, which are inherently weaker, reducing the effective strength. The effective tensile strength σ_{eff} for a certain raster angle θ can be modeled by a rule of mixtures

$$\sigma_{\text{eff}} = \sigma_{\parallel} \cdot \cos^2 \theta + \sigma_{\perp} \cdot \sin^2 \theta \quad (\text{Eq 10})$$

where $\sigma_{\parallel}$ is the applied strength in the filament loading direction, $\sigma_{\perp}$ is the applied strength in the filament bonding direction and θ is the angle of incidence of the raster with respect to the load direction. Physically, this formulation provides that for filaments arranged along the load axis, called helix ($\theta = 0^\circ$), the tensile strength is maximized because the load is carried by the continuous strands of polymers. At oblique orientation ($\theta = 45^\circ$) the stresses are shared between directions which gives a moderate tensile strength but a better crack deflection and resistance to fracture. At $\theta = 90^\circ$, the load

is primarily supported by weaker inter-filament bonds thus giving poor tensile performance.

In summary, raster orientation effects geometry and strength by the anisotropic shrinkage and load transfer pathways. Selecting the ideal angle involves a balance between dimensional stability and axon, and this is especially important on parts submitted to complicated or multi-diffuser loading conditions.

2.6 Nozzle Diameter (F)

Nozzle diameter is a fundamental parameter in FDM that determines the width and volume of each extruded filament strand. It has a direct influence on interlayer bonding, thermal diffusion, and print resolution, which is crucial for dimensional accuracy and mechanical performance.

Larger nozzles cause lower resolution, particularly in those features where there are fine curves and sharp corners. The dimensional fidelity is only constrained by the smallest printable feature of the nozzles, which is also inversely proportional to the diameter.

$$L_{\min} \approx k_r \cdot F \quad (\text{Eq 11})$$

where $L_{\min}$ represents the minimum printable feature size, k_r refers to the resolution coefficient (typically 1.2-1.5, depending on slicer settings). Smaller nozzles ($F < 0.40$ mm) improve the dimensional accuracy at the expense of inferior interlayer bonding due to thinner filaments and faster cooling rates providing reduced mechanical properties at an increased risk of porosity. Additionally, nozzle diameter also affects the Z height calibration. Discrepancies between bead width and layer thickness could introduce over or under extrusion, as well as the wall thickness and surface flatness.

On the contrary, a larger nozzle diameter results in wider filament roads, which provide increased contact area between adjacent extrusions and layers. This increases interfacial bonding strength, void formation, and increases the load-bearing capacity of the printed part. The cross-sectional area of the extruded bead, which determines the bonding interface, can be approximated as given as follows (Ref 32).

$$A_e = \frac{\pi}{4} \cdot F^2 \cdot \lambda \quad (\text{Eq 12})$$

where F is the diameter of the nozzle (in mm) and λ represents the expansion factor of the extruded material (empirically depending on the flow rate and viscosity of the material). Physically, this relation arises because bead width increases quadratically with nozzle size, providing more interface for stress transfer between layers. Stronger interfacial bonding improves load-bearing capacity and tensile strength. Higher A_e improves stress transfer between layers, thereby increasing tensile strength. However, there can be a point at which if the bead width is too wide, undulations on the surface and internal stresses from uneven cooling can be induced. Thus, an optimum nozzle diameter will ensure that the strongest parts and fastest part-to-part layer adhesion are produced with a wide nozzle diameter (e.g., 0.5 mm), whereas tight dimensional control and delicate detail resolution are created with a finer nozzle diameter (e.g., 0.3 mm). In dual-objective optimization, selecting nozzle diameters that maximize interfacial bonding while maintaining acceptable feature tolerance is essential for high-quality BB-PLA prints.

2.7 Extruding Speed (G)

Extruding speed—also referred to as print or deposition speed—dictates how fast the filament is fed through the nozzle and deposited on the build platform. It affects bonded part thermal history, bonding time and layer stability, hence also affecting part geometry tolerance and mechanical integrity. Reducing the speed can cause overheating, oozing, or material sag (especially in fine features) and thus loss of dimensional fidelity. On the other hand, too high speeds might lead to under extrusion, as well as bad bead placement and layer alignment, all of which would lead to surface defects and dimensional inaccuracy. A simple empirical model for extrusion-induced deviation ε_v as a function of speed can be expressed as

$$\varepsilon_v = \beta_1 v - \beta_2 \frac{1}{v} \quad (\text{Eq 13})$$

where ε_v is dimensional error (mm), β_1 and β_2 are material dependent and printer dependent constants, respectively. This relationship represents a nonlinear behavior of the relationship with both the high and low speeds resulting in errors an indication of the need for optimization of speed.

The extruded filament is kept hotter for longer at the slower speeds, allowing for better inter-layers thermal diffusion and better inter-layers bonding. Higher extrusion rates, however, reduce bonding time and may result in poor fusion because of premature cooling thereby reducing interfacial strength. The thermal contact time t_c between successive layers can be approximated as follows:

$$t_c = \frac{L}{v} \quad (\text{Eq 14})$$

where L refers to the filament path length per layer (mm) and v the extrusion speed (mm/s). For a given L , the lower v means a longer t_c , which leads to more heat being transferred to the preceding layer and the promotion of molecular entanglement—which is an essential requirement for tensile strength. This form is the direct result of kinematic considerations: slower speeds mean a greater residence time of the hot bead on the previous layer and the increased heat transfer, molecular diffusion and entanglement of the chains mentioned above which result in a greater interlayer adhesion and greater tensile strength. Conversely, high speeds shorten the contact time, limiting fusion and leading to weaker interfaces.

For BB-PLA, a moderate extrusion speed around 40–60 mm/s is optimally balanced between expert fusion time that

provides mechanical strength, controlled cooling time forming dimensional accuracy, and absence of thermally induced artifacts or lack of layer alignment. Thus, the extrusion speed should be kept in accordance with the nozzle diameter and temperature for stable material flow, mechanical stability, and dimensional stability.

3. Materials and Methods

This section describes the materials and methods used to identify the best process parameters to be used for 3D printing of BB-PLA.

3.1 Material

In this study, BB-PLA filament was used, which has been supplied by 3D Master India, a certified supplier of environment-friendly 3D printing material. No in-house synthesis and/or preparation of the BB-PLA was carried out, and thus the filament was used as received, which guaranteed the same material properties throughout the present study.

BB-PLA is an aliphatic polyester that is based on the use of renewable sources, e.g., corn or potato starch, and that is recognized for its biodegradability and favorable mechanical performance. Typical thermal and mechanical properties are below: melting point – 170 °C, density – 1.25 g cm⁻³, shear modulus – 2.4 GPa, tensile strength – 52 MPa. Table 1 summarizes the physical properties of the BB-PLA filament as per the manufacturer 3D Master India. In the context of this study, the thermal conductivity, specific heat, and thermal diffusivity of BB-PLA affect the thermal gradients during the printing process, strongly influencing the dimensional accuracy through the phenomenon of shrinkage and warping behavior. Simultaneously, various mechanical properties, such as tensile strength, elongation at fracture, and shear modulus governing its load-bearing capacity, are important in assessing tensile properties.

3.2 Equipment

BB-PLA specimens were fabricated using a Creality Ender-3 V3 printer. The printer used in this study is shown in Fig. 1. This printer has a power of 350W, build volume of 300 × mm300 × mm330 mm. It has a maximum print speed of 600 mm per s, and the temperature of the print nozzle is up to 300 °C. Although the Creality Ender-3 V3 is an open frame system, all experimental runs were done in a thermally controlled enclosure to have consistent ambient conditions and to minimize the external thermal gradient during the FDM process. In order to facilitate the replication of the experiment, Table 2 presents the essential printer configuration parameters used during the fabrication of BB-PLA specimens.

3.3 Methodology

This study adopts a systematic experimental and statistical methodology to evaluate the effect of key FDM process parameters on the dimensional accuracy and tensile strength of BB-PLA components. Figure 2 presents the systematic framework for optimizing FDM process parameters. The design of the investigation is based on the Taguchi design of experiments, here a L18 orthogonal array, which allows the efficient

Table 1 Physical properties of BB-PLA filament used in this study

Property	Value	Unit
Viscosity	0.26-0.46	M Ps-s
Thermal diffusivity	0.05	m ² /sec
Thermal conductivity	0.12	W/M°k
Density	1.25	g/cm ³
Fracture elongation	10-100	%
Flexural modulus	345-450	MPa
Specific heat	1800	J/Kg°K
Melting point	120-170	°C
Tensile strength	52	MPa
Shear modules	2.4	GPa

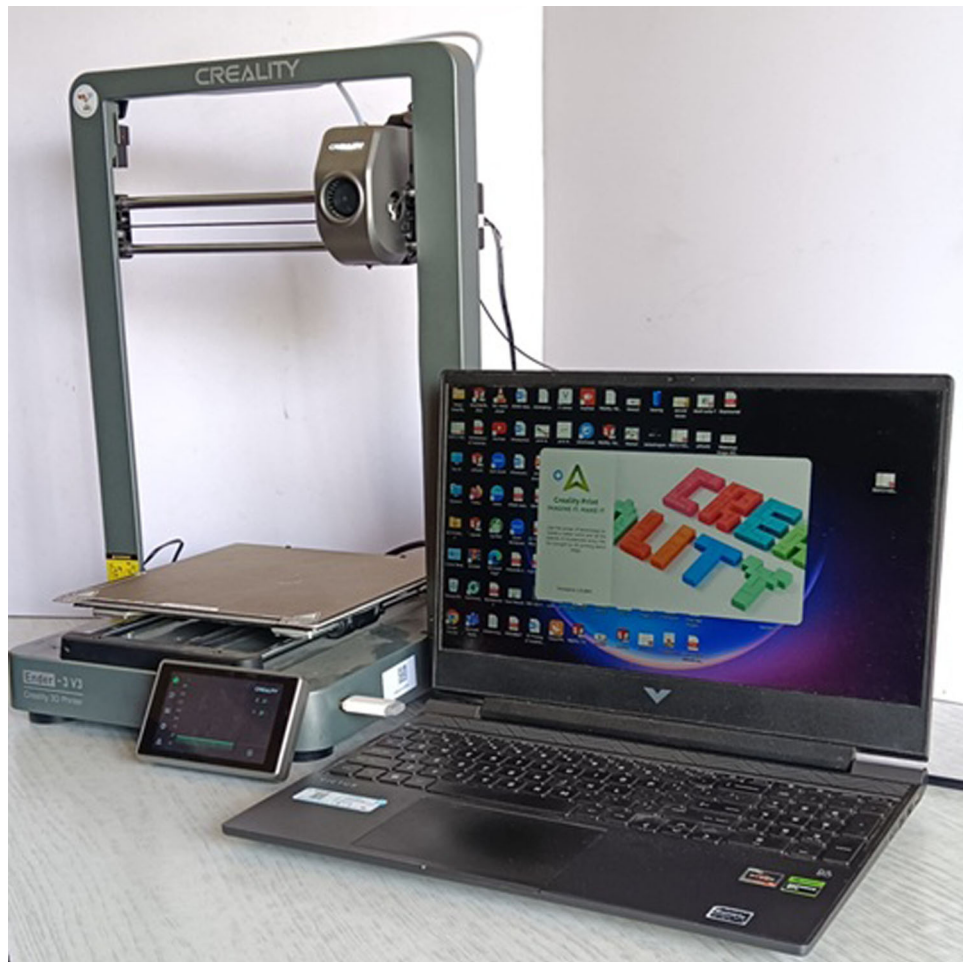


Fig. 1 3D printer used for FDM experiments

Table 2 Detailed FDM printing parameters used in the fabrication of BB-PLA specimens

Parameter	Value
Flow rate	100% ($\approx 4 \text{ mm}^3/\text{s}$)
Cooling fan percentage	100%
Retraction distance	5 mm
Retraction speed	25 mm/s
Bed temperature	60 °C
Brim	5 mm
Raft	Not used
Extrusion multiplier	1.00

investigation of multi-parameter combinations with reduced total numbers of experimental trials. Figure 2 shows the systematic framework for optimization of the FDM process parameters. An extensive literature review was first carried out for the identification of critical gaps in understanding the influence of FDM parameters on dimensional accuracy and mechanical performance, in particular tensile strength. Informed by past research and practice, seven key input parameters were chosen. These control parameters were varied according to predetermined levels to cover a region of high

response. Fabricated specimen's X , Y , Z dimensions and tensile strength were used as primary output response variables.

BB-PLA was chosen as the material because of its renewable origin, biodegradability, and its increasing industrial relevance in sustainably manufactured materials. All the specimens were designed following the ASTM standard and printed using a Creality Ender-3 V3 FDM printer. The slicing and the parameter control have been carried out using UltiMaker Cura software in order to maintain the consistency for all the experimental trials. After fabrication, three major linear dimensions were measured—length (X), width (Y), and thickness (Z)—using a digital Vernier caliper. A tensile test was then carried out at ambient temperature by using a universal testing machine with a load cell of 20 tons with a loading accuracy of $\pm 1\%$. The speed of the test was held constant at 5 mm min^{-1} . In order to assure the statistical reliability, three replicates were performed in each of the experimental conditions. Data analysis was based on the calculation of Signal-to-Noise (S/N) ratios to determine parameter settings that maximize the dimensional accuracy and tensile strength and then using the criterion “bigger is better” for selection. Moreover, Analysis of Variance (ANOVA) was carried out to determine the statistical significance of each parameter and for the quantification of the contribution of each factor to the tensile response. A multiple linear regression was developed to

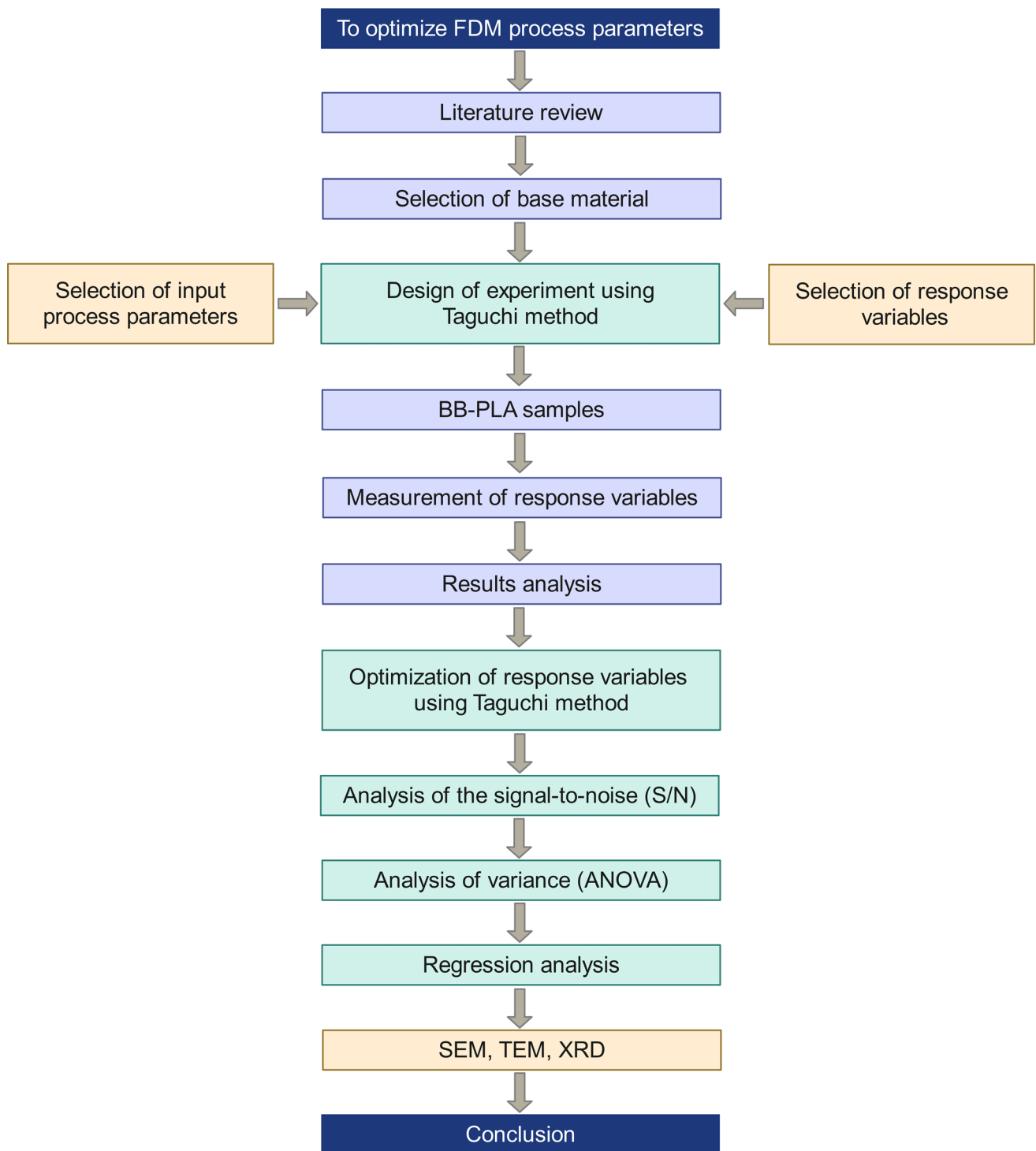


Fig. 2 Systematic framework for optimizing FDM process parameters

characterize the relationship between tensile strength and the studied process variables.

In addition to dimension and mechanical analyses, samples were studied using scanning electron microscopy and x-ray diffraction to examine their fracture surfaces and determine the degree of crystallinity. Such microstructural evaluations are critical for understanding how internal features translate into macroscopic mechanical performance in polymer-based additive manufacturing. At the end of the methodology, the best

parameter combination was determined and checked by means of empirical validation. The combined results were disseminated in guidelines for optimization of dimensional accuracy and tensile strength of FDM-produced BB-PLA parts with practical value and scientific contribution to the worldwide field of sustainable additive manufacturing.

Figure 3 shows the integrated workflow and the set of software tools used for the design, fabrication, and analytical phases of the fabrication process of BB-PLA specimens as

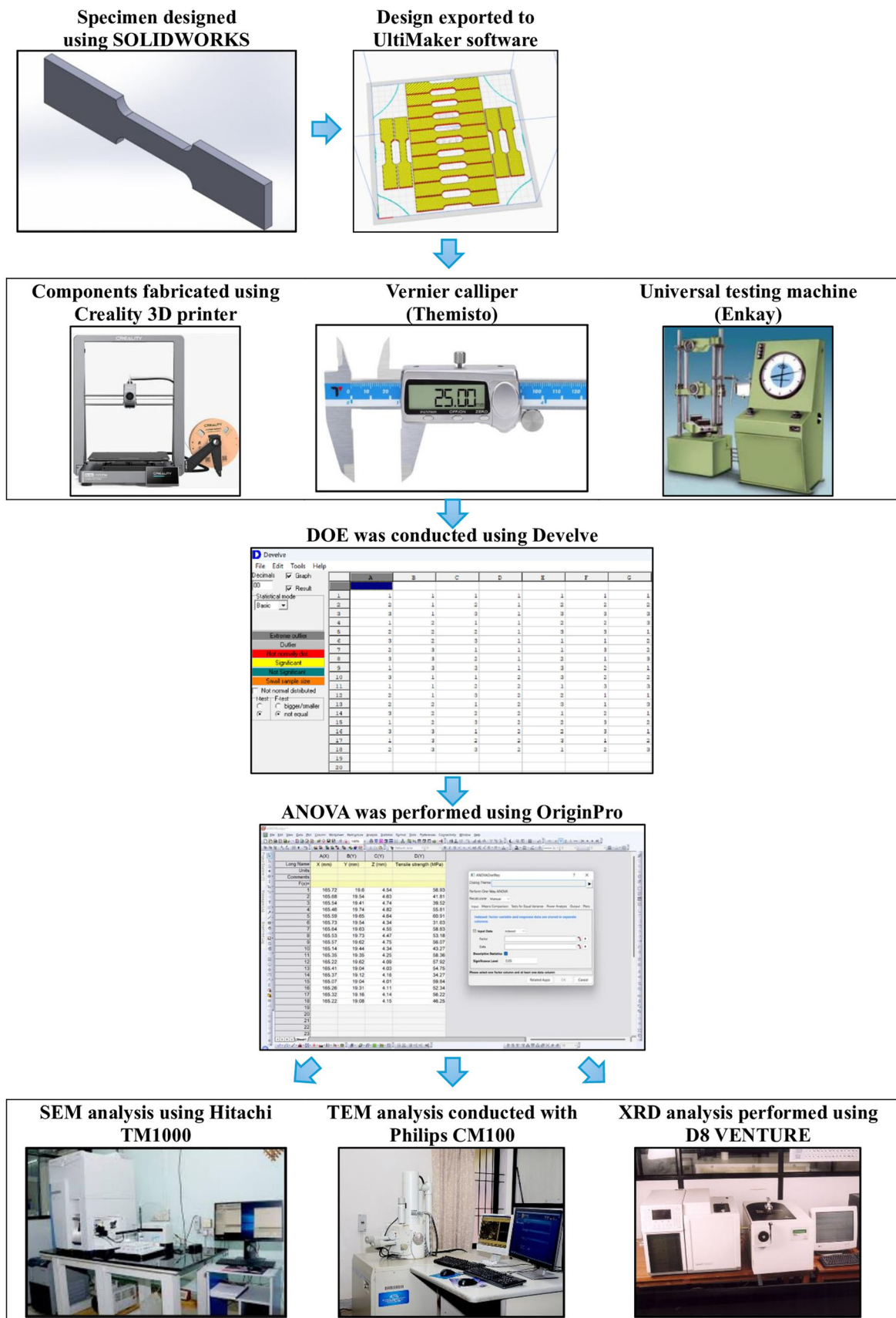


Fig. 3 Comprehensive workflow and tools for specimen design, fabrication, and analysis in the study

developed in this study. A three-dimensional model of the tensile specimen geometry was created using SolidWorks, which meets the ASTM specification, and the CAD file was exported to UltiMaker Cura, a three-dimensional slicing software for Creality Ender-V3 3D printer. This software was used to apply different FDM process parameters and was used to generate G-code for subsequent fabrication. Specimen dimensions were measured by using Themisto digital Vernier caliper, and material tensile tests were conducted using a 20-ton capacity universal testing machine (Enkay Enterprises).

Experimental method and parameter variation were controlled through Develve software by means of the Taguchi L18 orthogonal array in order to perform a systematic and statistically efficient search toward the parameter space. OriginPro was used for statistical analysis and to calculate S/N ratios, which aided in determining the importance of parameters and their contribution to the variance in tensile strength. Scanning Electron Microscopy (SEM) was used to observe fracture surface morphology; Transmission Electron Microscopy (TEM) was used to observe internal microstructure; x-ray diffraction (XRD) was used on a Philips TEM to determine the crystallinity and phase structure of the printed BB-PLA samples.

3.4 Selection of Process Parameters and Their Levels

Process parameters and their levels for this study were chosen based on a comprehensive literature review and inspired by the work of Hikmat et al. (Ref 33), who have already shown that the printing conditions have a vital role in the tensile properties of PLA components prepared by the FDM process. To systematically evaluate their effects, seven key process parameters were selected based on their known and hypothesized impact on dimensional accuracy and mechanical strength, material consolidation, and thermal behavior during the deposition process. These parameters are infill percentage, layer thickness, material temperature, build orientation, raster orientation, nozzle diameter, and extruding speed. Every parameter was set at three levels, except build orientation, which was assigned at two levels, so the Taguchi L18 orthogonal array design could explore their individual and combined impacts. For each parameter, the three levels were selected according to the process range as recommended by the filament manufacturer, other published works (Ref 33-35), and preliminary trial experiments that will prevent print failure and dimensional uniformity.

Infill percentage (A): it controls internal density and load bearing capacity. Infill is increased to usually increase mechanical strength at the expense of material usage. Layer thickness (B): Layer thickness affects inter-layer bonding and surface finish; thinner layers may improve bonding, but will lengthen

Table 3 Process parameters and their levels

Parameters	Level 1	Level 2	Level 3
Infill percentage (A)	80%	90%	100%
Layer thickness (B)	0.2 mm	0.3 mm	0.4 mm
Material temperature (C)	190 °C	200 °C	210 °C
Build orientation (D)	X (flat)	Y (on-edge)	...
Raster orientation (E)	0°	30°	45°
Nozzle diameter (F)	0.3 mm	0.4 mm	0.5 mm
Extruding speed (G)	20 mm/s	40 mm/s	60 mm/s

print time. Material temperature (C) has an impact on the flow of the filaments, the adhesion between each layer and the thermal fusion. The selected range assures the processing on the thermal stability range of BB-PLA. Build orientation (D) significantly affects load distribution and interlayer bonding, with on-edge and flat orientations providing contrasting stress responses. Raster orientation (E) is a measure of the direction of filament paths relative to the applied load and can modulate crack propagation behavior. Filament width is determined by a nozzle diameter (F) that will determine the resolution and heat-transfer properties of the print. Extruding speed (G) affects deposition consistency and the rate of solidification, influencing mechanical performance and surface integrity. Each parameter was varied within a reasonable and industry-common range, thereby allowing the data to be used for broader applications in addition to providing the potential for statistical analysis of tensile strength.

3.5 Design of Experiments

The effect of seven FDM process parameters which significantly affect the dimension accuracy and tensile strength of BB-PLA specimens was systematically evaluated using a Taguchi array-based design of experiment. The Taguchi method is particularly suitable for optimizing multi-factor systems with minimal experimental effort, offering robust insights into parameter effects and interactions while reducing variability due to uncontrollable external influences.

As mentioned above, seven process parameters were chosen including infill percentage (A), layer thickness (B), material temperature (C), build orientation (D), raster orientation (E), nozzle diameter (F), and extruding speed (G), all of the three defined levels, except for build orientation that had two levels (flat and on edge). The levels were selected based on material manufacturer recommendations, literature precedent, and pilot tests to ensure printability and mechanical integrity as indicated in Table 3. The percentage infill of the samples was selected uniquely at 80%, 90%, and 100% in order to account for the range of transition from relatively lightweight but partially filled structures to fully solid parts, so that the effect of internal

Table 4 Taguchi orthogonal array L18 (coded parameters)

Trial	A	B	C	D	E	F	G
T1	A1	B1	C1	D1	E1	F1	G1
T2	A2	B1	C2	D1	E2	F2	G2
T3	A3	B1	C3	D1	E3	F3	G3
T4	A1	B2	C1	D1	E2	F2	G3
T5	A2	B2	C2	D1	E3	F3	G1
T6	A3	B2	C3	D1	E1	F1	G2
T7	A2	B3	C1	D1	E1	F3	G2
T8	A3	B3	C2	D1	E2	F1	G3
T9	A1	B3	C3	D1	E3	F2	G1
T10	A3	B1	C1	D2	E3	F2	G2
T11	A1	B1	C2	D2	E1	F3	G3
T12	A2	B1	C3	D2	E2	F1	G1
T13	A2	B2	C1	D2	E3	F1	G3
T14	A3	B2	C2	D2	E1	F2	G1
T15	A1	B2	C3	D2	E2	F3	G2
T16	A3	B3	C1	D2	E2	F3	G1
T17	A1	B3	C2	D2	E3	F1	G2
T18	A2	B3	C3	D2	E1	F2	G3

pore structure could be ascertained with respect to both strength and dimensional stability. Three different layer thicknesses of 0.20 mm, 0.30 mm, and 0.40 mm were chosen to make up the trade-off between surface quality, build speed, and interlayer adhesion, as thicker layers are advantageous to deposition efficiency and thin layers to accuracy. The processing temperature window of BB-PLA is from 190 °C to 210 °C, with lower processing temperatures leading to more stable results and higher processing temperatures leading to more inter-layer diffusion. The midpoint temperature of 200 °C gives equal measure between these effects. Orientation was only established as flat (X) and on-edge (Y) as these give the most contrasting expressions of anisotropy, where flat is favored for dimensional precision and on-edge for preloading of tensile load propagation. Moreover, 0°, 30°, and 45° raster orientations were added in to determine the effects of the filament alignment and cross-hatching on the stress distribution and crack propagation. Three different nozzle diameters of 0.30 mm, 0.40 mm, and 0.50 mm were used to represent the range from small-scale deposition reliability to higher bead widths for improved retention and bond. Finally, a series of extruding speeds of 20 mm s⁻¹, 40 mm s⁻¹, and 60 mm s⁻¹ was selected to investigate the trade-off between bonding time and production within the optimum range, while lower extruding speed resulted in high strength improvement by optimal fusion, while faster extruding speed was challenged for deposition stability in rapid manufacturing.

An L18 orthogonal array was adopted as the experimental matrix to accommodate this mixed-level factor design. The L18 orthogonal array was selected due to its suitability for accommodating mixed-level factors while minimizing the number of required experimental runs. The array balances parameter combinations while maintaining statistical efficiency, requiring only 18 experimental runs instead of the 2,187 (3⁶ × 2¹) trials needed for a full factorial design. This approach saves greatly on the experimental workload while making it possible to make meaningful analysis. Table 4 contains the full coded parameter settings for all the 18 trials. Each row represents a unique combination of the selected levels of the selected parameters used to fabricate one tensile specimen, which was tested under identical conditions. The overall analysis is based on formal effects and identifies the best combinations of the parameters. Statistical analyses (S/N ratio calculation and ANOVA) were performed to evaluate the contribution of each factor to dimensional accuracy responses and tensile strength responses and also to identify statistically significant influences suggested by Taguchi methodology.

The ASTM D638-I specification was adopted for specimen geometry to ensure comparability to existing literature regarding FDM printed components. This standard is used extensively in the measurement of dimensional and mechanical properties of polymer-based materials produced by additive manufactur-

ing. The test specimens for use in this study had dimensions of 165 mm × 19 mm × 4 mm. Accurate length, width, and height of specimens were measured by digital Vernier caliper for proper dimensional accuracy evaluation.

4. Experimental Results and Discussion

Specimen preparation was done on a 3-D printer at a controlled ambient temperature (T_a) of 24 °C. Figure 4 shows the dimensional configuration of the test specimen designed in compliance with ASTM D638 (I) standard. Figure 5(a) shows the manufactured specimens, each of which was made with different combinations of seven critical FDM process parameters as defined in the experimental design. Figure 5(b) shows the fabricated BB-PLA specimens in terms of their corresponding building orientations as used during the FDM print process.

4.1 Dimensional Accuracy Evaluation

The optimization of FDM process parameters with respect to the improvement in dimensional accuracy and tensile strength of BB-PLA components was performed by using the Taguchi method. Eighteen experimental runs were conducted according to the L18 orthogonal array, and the dimensional measurement and tensile strength results are summarized in Table 5. Dimensional accuracy was measured for FDM printed BB-PLA parts using three-dimensional measurements (X, Y, and Z), and two important metrics were calculated: average dimensional error (%) and maximum single-axis deviation (mm). These measures will give a full understanding of the geometrical fidelity of the printed parts.

4.1.1 Trial-Wise Accuracy Overview. Figure 6 presents the average dimensional error for all 18 trials. Errors varied from 0.16 percent (T15) to 8.22 percent (T4). Thus, wide variability occurred that depended on process settings. Trial T15, with the highest tensile strength (59.84 MPa), displayed better geometric conformity with less deviation (0.07 mm), thus proving that it is possible to obtain both strength and precision simultaneously. In contrast, trial T4, although being strongly performed, showed low accuracy (8.22%), which implies suboptimal geometrical control.

Further, some trials involved high variability in tensile strength, with high values of standard deviation. Noteworthy to mention, trials T3, T6, T13, and T16 had somewhat high standard deviations of tensile strength measurements, meaning greater variability in mechanical response. This variability has been attributed to a combination of suboptimal parameter interactions. Specifically, trial T3 used the largest nozzle diameter (0.50 mm), greatest temperature (210 °C), and greatest infill, which may result in local overheating and nonuniform bonding as a result of rapid thermal cycling. Trial T6 was done

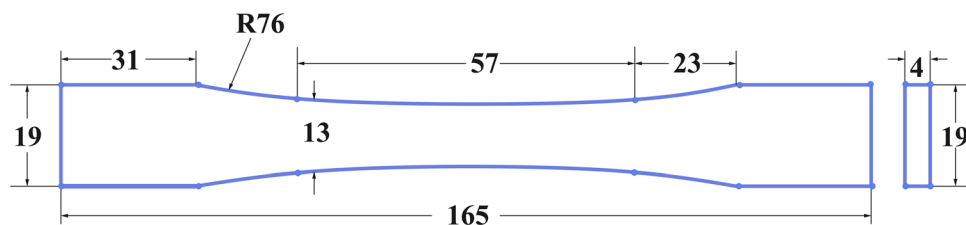


Fig. 4 Dimensions of the specimen based on ASTM D638-I standard (all dimensions are in mm)

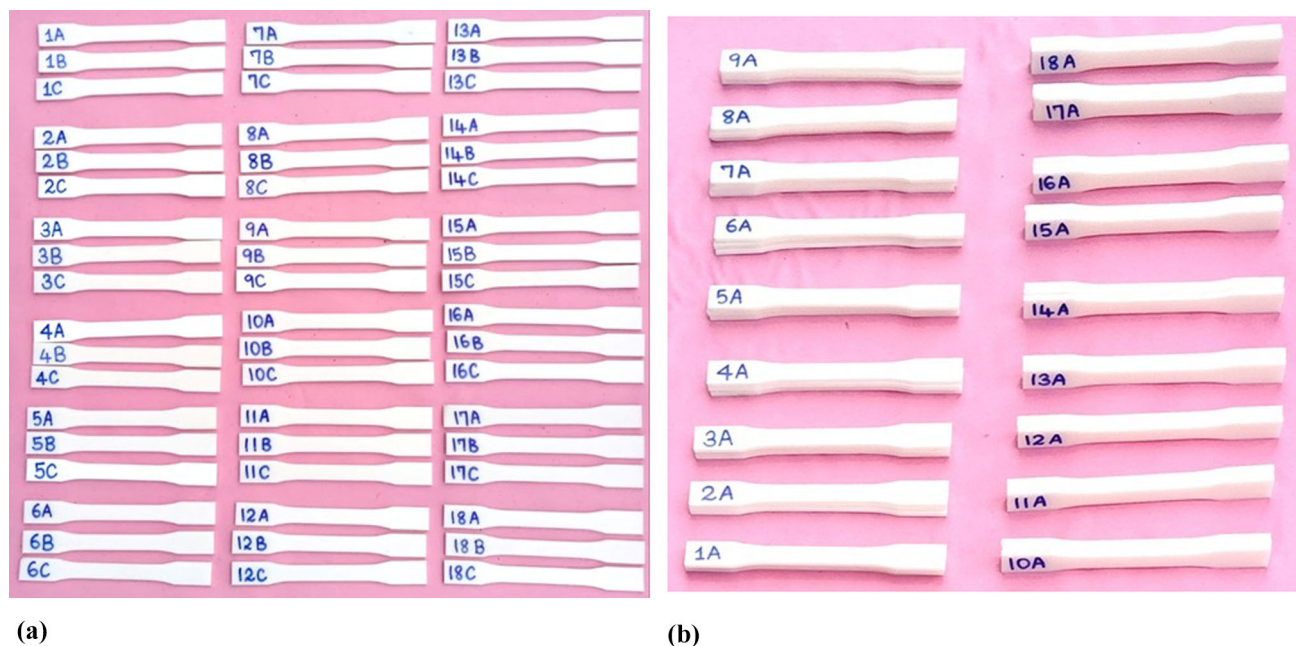


Fig. 5 Preparing specimens through additive manufacturing process using BB-PLA (a) Fabricated specimens based on seven FDM process parameter variations (b) Fabricated BB-PLA specimens and their corresponding build orientations

Table 5 Experimental results for combined evaluation of dimensional accuracy and tensile strength of FDM-printed BB-PLA specimens

Trial	A	B	C	D	E	F	G	Dimension			Average dimensional error, %	Maximum deviation, mm	Mean tensile strength, MPa	S/N ratio, dB	Standard deviation
								X, mm	Y, mm	Z, mm					
T1	A1	B1	C1	D1	E1	F1	G1	165.72	19.60	4.54	5.69	0.72	56.93	35.20	0.78
T2	A2	B1	C2	D1	E2	F2	G2	165.68	19.54	4.63	6.33	0.68	41.81	32.72	0.57
T3	A3	B1	C3	D1	E3	F3	G3	165.54	19.41	4.74	6.99	0.74	39.52	31.38	3.22
T4	A1	B2	C1	D1	E2	F2	G3	165.46	19.74	4.82	8.22	0.82	55.81	34.47	0.72
T5	A2	B2	C2	D1	E3	F3	G1	165.59	19.65	4.64	6.59	0.65	60.91	35.17	0.81
T6	A3	B2	C3	D1	E1	F1	G2	165.73	19.54	4.34	3.92	0.73	31.63	29.11	2.92
T7	A2	B3	C1	D1	E1	F3	G2	165.64	19.63	4.55	5.81	0.64	58.83	35.16	0.57
T8	A3	B3	C2	D1	E2	F1	G3	165.53	19.73	4.47	5.30	0.73	53.18	34.35	1.12
T9	A1	B3	C3	D1	E3	F2	G1	165.57	19.62	4.75	7.45	0.75	56.07	35.34	0.67
T10	A3	B1	C1	D2	E3	F2	G2	165.14	19.44	4.34	3.63	0.44	43.27	32.56	0.52
T11	A1	B1	C2	D2	E1	F3	G3	165.35	19.35	4.25	2.76	0.35	58.36	35.26	1.93
T12	A2	B1	C3	D2	E2	F1	G1	165.22	19.62	4.09	1.88	0.62	57.92	35.16	1.65
T13	A2	B2	C1	D2	E3	F1	G3	165.41	19.04	4.03	1.58	0.41	54.75	34.09	2.48
T14	A3	B2	C2	D2	E1	F2	G1	165.37	19.12	4.16	1.61	0.37	34.27	30.46	1.65
T15	A1	B2	C3	D2	E2	F3	G2	165.07	19.04	4.01	0.16	0.07	59.84	35.72	0.81
T16	A3	B3	C1	D2	E2	F3	G1	165.26	19.31	4.11	1.51	0.31	52.34	34.15	3.82
T17	A1	B3	C2	D2	E3	F1	G2	165.32	19.16	4.14	1.51	0.32	56.22	34.87	0.56
T18	A2	B3	C3	D2	E1	F2	G3	165.22	19.08	4.15	1.43	0.22	46.25	33.35	0.84

with a smaller nozzle (0.30 mm) and also high temperature (210 °C), but flat build orientation may cause adequate under-extrusion and under-bonding due to fast cooling.

Besides, trial T13 showed relatively high standard deviation (2.48 MPa), probably due to the combination of small nozzle diameter (0.30 mm) and high extruding speed (60 mm s⁻¹) with low material temperature (190 °C). These conditions had the potential for creating insufficient extrusion, inadequate inter-layer bonding, and thermal unevenness. The slope of the fillets on the intake in the 45° raster direction might have

additionally contributed to the stress anisotropy and unequal transfer of load, and the 90% infill might have created internal discontinuities. As a result of these factors, the ultimate tensile behavior is expected to be specimen dependent. Trial T16 was the thickest (thickness = 0.40 mm), slowest (speed = 20 mm s⁻¹) and with high infill (100%), which would be the reason for localized overheating, low cooling efficiency, and uneven bonding due to long thermal residence time, which further developed thermal stress and porosity, leading to the variability in tensile strength.

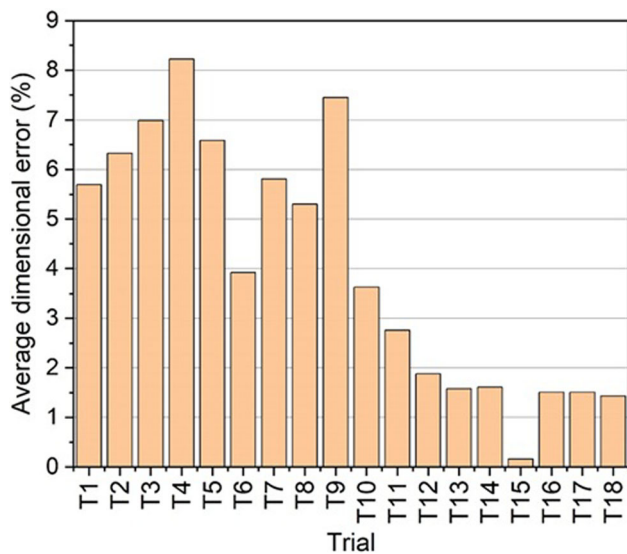


Fig. 6 Average dimensional error (%) across all experimental trials

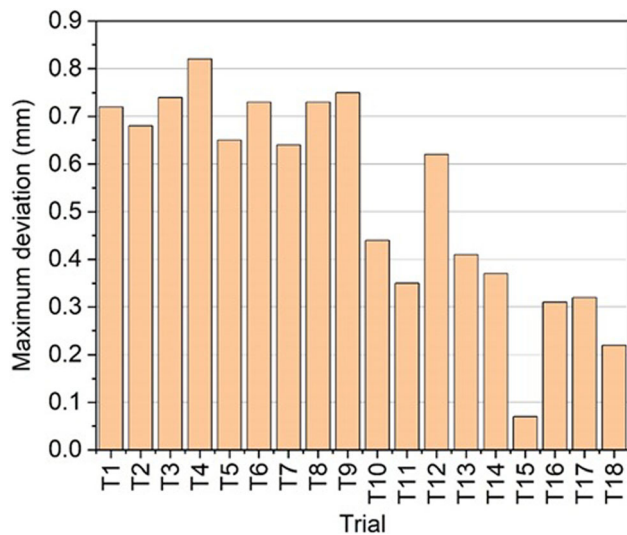


Fig. 7 Maximum single-axis deviation (mm) observed per trial

4.1.2 Deviation Sensitivity. As shown in Fig. 7, the maximum error in single axis direction was up to 0.82 mm in T4, confirming that the single-dimensional offsets alone can greatly degrade accuracy. Such tolerances are very important for applications needing high tolerance levels. Again, the best geometry (0.07 mm) was found for T15, confirming its position as a dual-optimization candidate.

4.1.3 Accuracy-Strength Trade-Off. Figure 8 shows the correlation between average dimensional error and tensile strength. The data show no hard and fast inverse relationship and, in essence, confirm that the process of tensile enhancement need not result in deteriorating dimensional accuracy. T15 showed good performance in both low-dimensional error (3.92%) and strength (31.63 MPa) compared to other trials (e.g., T6), which showed poor performance in strength. As shown in Table 4, the maximum tensile strength of 59.84 MPa occurred in trial T15, while the minimum tensile strength of 31.63 MPa was found after T6. This significant variation

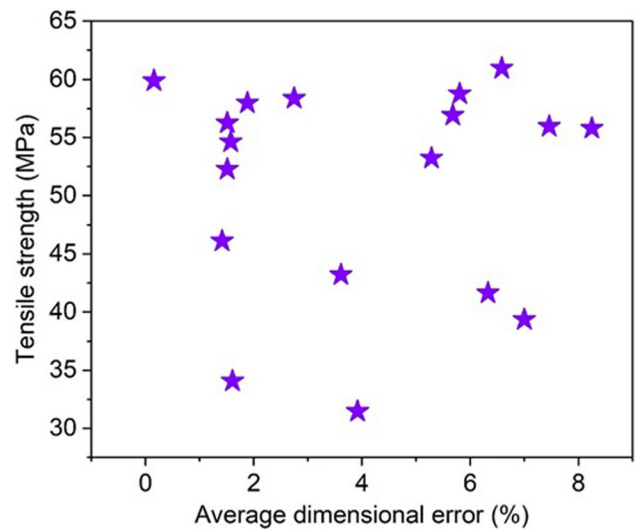


Fig. 8 Correlation between dimensional accuracy and tensile strength

highlights the considerable impact that process parameters have on the dimensional accuracy and mechanical behavior of FDM-fabricated parts.

4.2 Optimal Process Parameters

To select the best parameters combinations for optimizing both dimensional accuracy and tensile strength, the signal-to-noise ratio (larger the better criterion) of each run was calculated.

$$S/N = -10 \log \left[\frac{1}{n} \sum_{i=1}^n \frac{1}{Y_i^2} \right] \quad (\text{Eq 15})$$

This formula is suitable when the S/N ratio is inversely proportional to the geometrical error, that is, the higher the geometric error, the lower the S/N ratio, while both mechanical properties such as tensile strength and the deformation form are maximized while geometric error is minimized. In this study, $n = 1$ since the mean of the three replicate tests was used to represent each experimental run. The mean S/N ratios of all process parameters for each level are summarized in Table 6. The results of the analysis showed that the best combination is infill of 80%, layer thickness of 0.40 mm, material temperature of 190 °C, and on-edge build direction. Other conditions that also were favorable for geometric fidelity included a raster orientation angle of 45°, a nozzle diameter of 0.50 mm, and an extruding speed of 20 mm/s, all of which are known to minimize thermal gradients and to provide greater dimensional reproducibility.

The main-effect plot of signal-to-noise ratio (Fig. 9) confirmed that build orientation, nozzle diameter, and infill percentage have the most significant effect on the printing performance. These parameters have a critical impact both on mechanical strength, as a result of thermal bonding and filament overlapping, as well as on dimensional accuracy, as a result of controlling heat flow and layer alignment. In contrast, the effects of material temperature and the orientation of the raster do have a relatively lesser influence within the values that have been tested, which suggests that these influences are more nuanced and secondary. The results support the potential of

Table 6 Level-wise S/N ratio summary for FDM process optimization

Levels	Infill percentage (A)	Layer thickness (B)	Material temperature (C)	Build orientation (D)	Raster orientation (E)	Nozzle diameter (F)	Extruding speed (G)
Level 1	33.38	32.73	33.16	31.76	32.52	31.57	33.34
Level 2	32.97	33.25	32.64	34.25	33.32	33.27	33.27
Level 3	32.64	33.81	32.76	...	33.45	33.37	32.54
Delta	1.66	0.94	0.24	2.16	0.25	1.67	0.33
Ranking	3	4	6	1	7	2	5

Bold values indicate the optimal combination of process parameters

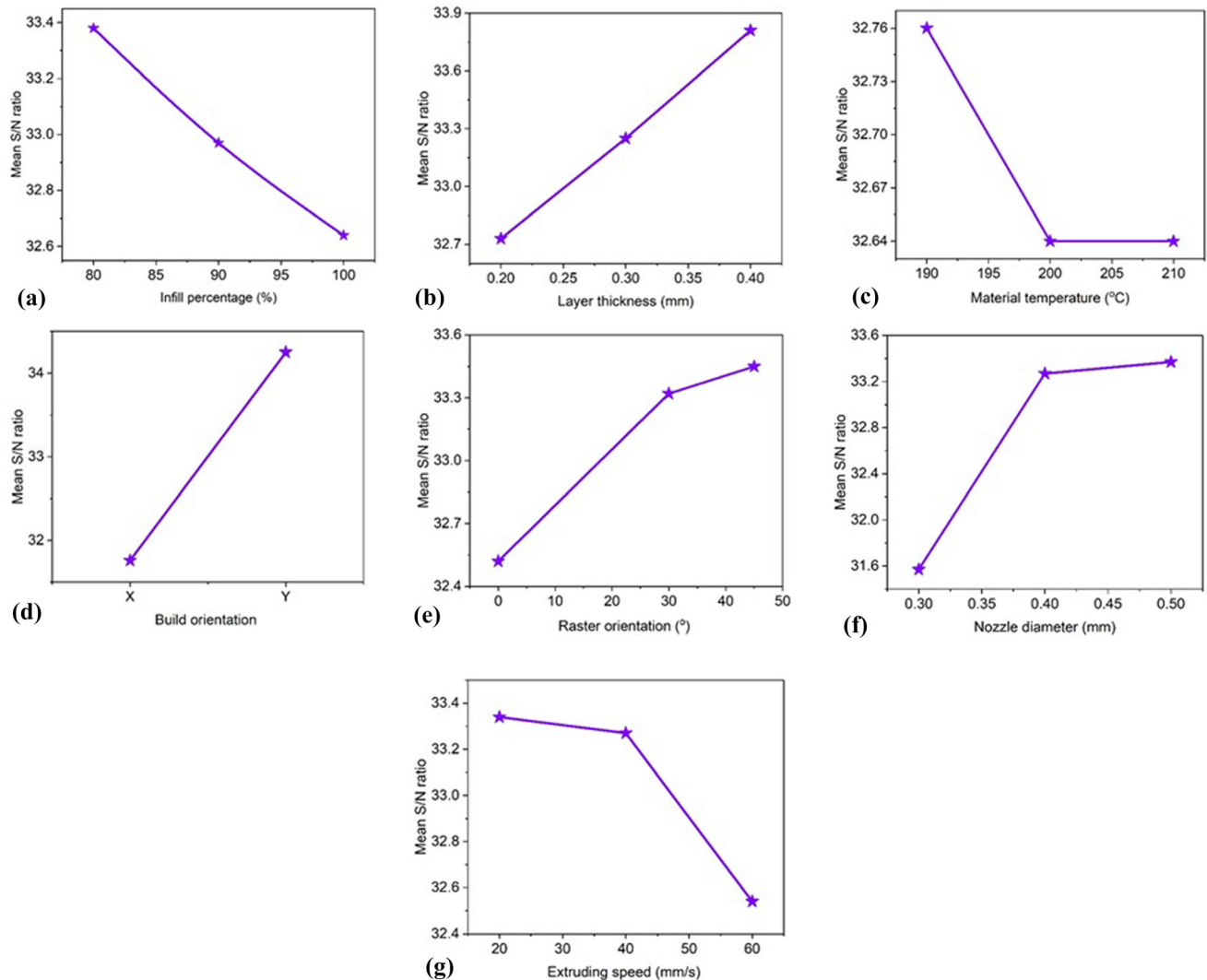


Fig. 9 Main effects plots of S/N ratios across FDM parameter levels, highlighting their influence on both dimensional accuracy and tensile strength of BB-PLA parts (a) Infill percentage (b) Layer thickness (c) Material temperature (d) Build orientation (e) Raster orientation (f) Nozzle diameter (g) Extruding speed

multi-objective optimization in FDM; worthwhile combinations of parameters (e.g., high infill coupled with moderation in layer thickness and stable extrusion) can be used to pursue both improved geometrical accuracy and tensile strength of prints in BB-PLA.

Figure 10 shows the stress–strain plots for specimens with the minimum and maximum tensile strengths for respective build orientations. The sample printed along the longitudinal direction presents a steep elastic zone followed by a brittle

fracture with a small plastic deformation, as is considered brittle behavior, as shown in Fig. 10(a). This response indicates a high inter-layer bonding strength, that energy is dissipated during the deformation stage and does not crack until the material reaches its tensile limit.

In Fig. 10(b), the stress–strain response of the specimen fabricated in the flat orientation is shown. The curve was obtained beyond the linear elastic region, which showed an enlarged region of plastic deformation prior to crack formation.

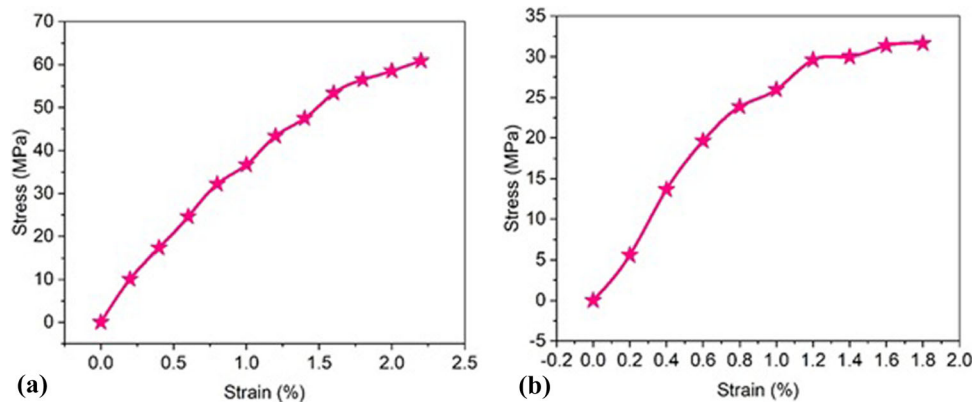


Fig. 10 Comparison of stress–strain behavior at extreme tensile strength conditions (a) Maximum value (Trial T15) (b) Minimum value (Trial T6)

Table 7 ANOVA results for tensile strength based on FDM parameters

Parameters	DOF	SS	V	F-test	P%	P value
Infill percentage (A)	2	10.53	5.27	10.35	18.98	0.019
Layer thickness (B)	2	4.72	2.36	4.63	8.51	0.063
Material temperature (C)	2	0.46	0.23	0.45	0.83	0.652
Build orientation (D)	1	23.35	23.35	45.78	42.07	0.004
Raster orientation (E)	2	0.47	0.24	0.47	0.85	0.639
Nozzle diameter (F)	2	13.37	6.69	13.12	24.09	0.012
Extruding speed (G)	2	0.53	0.27	0.53	0.96	0.608
Residual error	4	2.05	0.51	...	3.69	...
Total	17	55.48	...	...	100	...

The specimen shows a more ductile behavior probably due to weaker inter-layer bonding in the *X–Y* plane, which helps the specimen to stretch more before failure. And, though the overall tensile strength is less, the material has a greater capacity to deform more before fracture.

4.3 Analysis of Variance

To statistically check the effects of all process parameters on dimensional accuracy and tensile strength, an analysis of variance (ANOVA) was carried out on the experimental results as shown in Table 7. The ANOVA revealed statistically significant three parameters (build orientation, nozzle diameter, and infill percentage) at the 95% confidence level ($p < 0.05$), and these parameters were proved to be significant in deciding the mechanical performance of BB-PLA parts that are fabricated using FDM. These parameters also showed conceivable effects on dimensional error indexes, highlighting the dual functionality responsible for the monitoring of part quality. Similar trends have been observed in previous studies, notably in Ref 33, 35, 36, supporting the validity of the present results.

Build orientation was found to be the most significant factor, accounting for 42.07% of the variation in tensile strength ($p = 0.004$). Print parts in the on-edge position (*Y*-axis) showed improved bonding between layers (along with the direction of the applied stress) that increased load-transfer efficiency and decreased warping in the *Z*-axis, leading to improved dimensional accuracy. Nozzle diameter had an effect on strength variation of 24.09% ($p = 0.012$). The 0.5 mm nozzle diameter suggests wider paths for depositing the filaments that help in

better thermal fusion, and it also lowers the air void, thereby increasing the geometrical fidelity of the wide paths of deposition, helping uniform deposition of layers and reducing irregularities on the surface, which in turn will play a major role in maintaining tight dimensional tolerances. Infill percent contributed 18.98% ($p = 0.019$) and affected significantly both parameters. Infill percentage of 80% showed improved mechanical integrity combined with an internal deformation reduction, resulting in the maintenance of the preferred geometrical profile when using appropriate thermal and mechanical loads.

On the other hand, the other parameters such as layer thickness, material temperature, orientation of the raster, and extrusion speed did not reach statistical significance within the range of values studied, but layer thickness came close to it ($p = 0.063$), which indicates a secondary influence that can be stronger under different conditions. In conclusion, the ANOVA shows that optimization of the build orientation, nozzle diameter, and infill percentage are key factors to improve dimensional accuracy and tensile strength of FDM printed BB-PLA parts.

Figure 11 shows the tensile stress behavior of the BB-PLA and the specimens produced from a range of FDM process parameters. Of these, build orientation was found to be dominant. In Fig. 11(d), it can be seen that the tensile strength of the specimens with on-the-edge direction is increased in comparison to the flat direction, which showed an increment of over 10 % in comparison. This increase is due to the increased inter-layer bonding strength in the *Z*-direction, which acts as the main load-bearing mechanism when the part is placed on

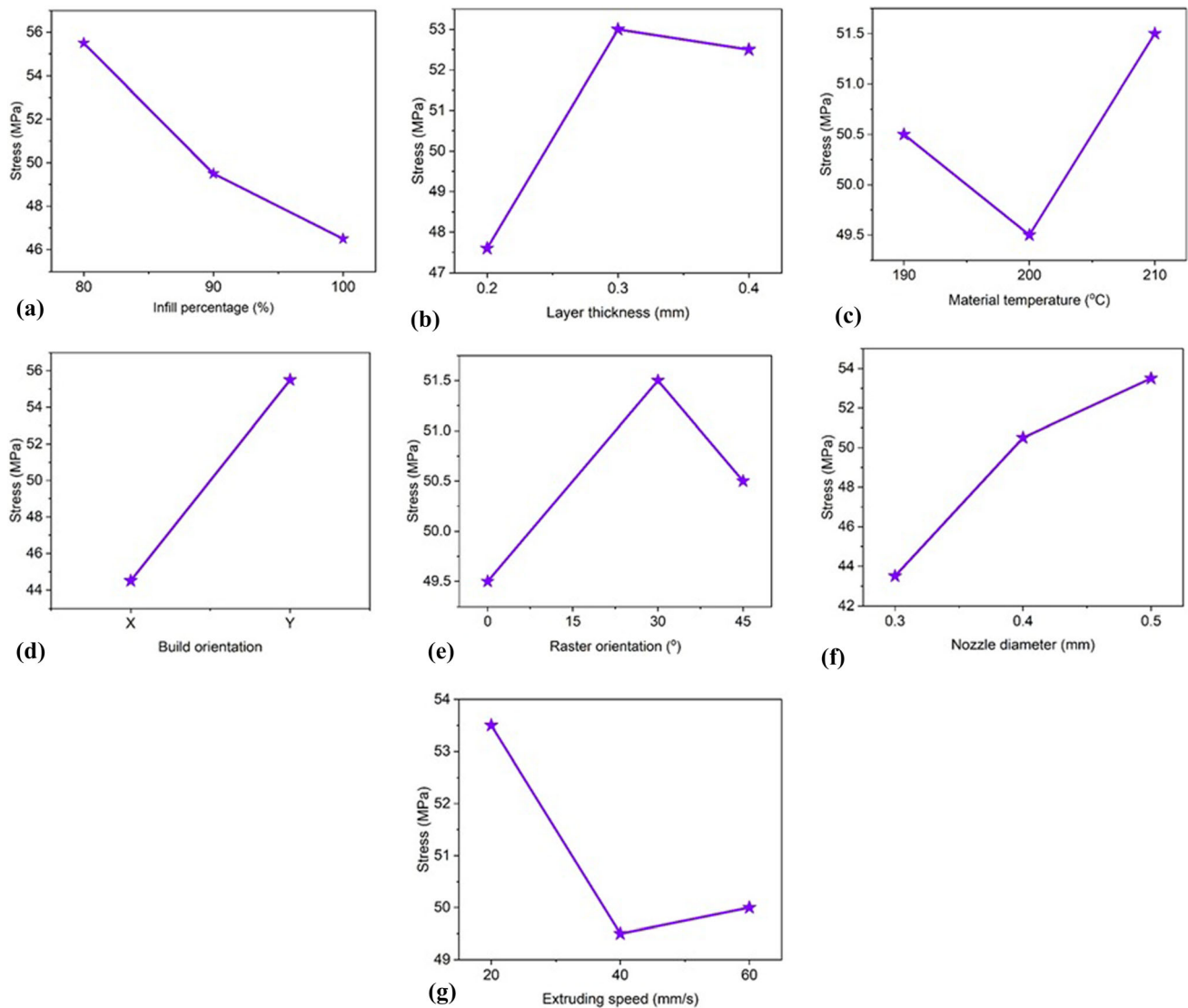


Fig. 11 Comparison of tensile stress curves across process parameters (a) Infill percentage (b) Layer thickness (c) Material temperature (d) Build orientation (e) Raster orientation (f) Nozzle diameter (g) Extruding speed

edge. In this configuration, the greater number of layers and improved interface parallelism between the inter-layers and load results in improved stress transfer and structural cohesion.

On the other hand, in the flat setting, the specimen has fewer layers and a wider cross-sectional area and thus the loading is mostly along trans-layer bonds in the X–Y plane. These bonds tend to be weaker because of the low level of filament interaction as well as inefficient heat retention between the paths deposited. In addition, the small geometry of the on-edge specimen (4 mm wide) helps to preserve the thermal energy from layer to layer, thus improving the bonding by thermal diffusion. However, when increasing Z- the part thickness without increases in width, more internal defects may be expected, such that the Z- thickness improvement may not provide mechanical strength.

Nozzle diameter is the second most dominating parameter, contributing 24.09% to tensile strength variation. As shown in Fig. 11(f), when the nozzle diameter is increased from 0.3 mm to 0.5 mm, the tensile strength increases significantly, from 43.5 MPa to 53.5 MPa. The major reason for this improvement

is to be attributed to the enlarged deposition lines as a result of a wider deposition nozzle opening, leading to additive improvement in the correlation with descent road and better adhesion between adjacent paths. Larger diameter filaments also stay hot longer, which helps the diffusion of the molecules and creates stronger welds. In semi-crystalline polymers such as PLA, slow cooling makes it easier for high crystalline zones that offer superior mechanical performance to form as opposed to amorphous zones. Thus, greater nozzle diameter yields enhanced crystallinity and, thus, greater tensile performance. Infill percentage also has a significant influence on tensile behavior, Fig. 11(a). At 80% infill, the specimen is virtually void-free, the structure is dense and continuous, and can resist loads applied to it more efficiently. Larger infill avoids internal voids, thus allowing for an increasing load spread over a wider cross-sectional area. Second, a reduction in the amount of air between adjacent filaments also reduces convective heat loss to allow for hotter temperatures to be maintained during deposition. This thermal retention improves inter-layer adhesion, crystallinity, and strength. Conversely, with lower infill densi-

ties, structural discontinuities are allowed, due to the increase in the risk of stress concentration.

4.4 Combined Influence of Key Process Parameters

While the tensile strength is highly dependent on the individual process parameters such as build orientation, nozzle diameter, and infill percentage, the interactions between these parameters prove to be of similar importance for the dimensional accuracy as well as mechanical performance of BB-PLA parts. The results indicate that optimal part quality is not obtained through isolated environments but through well-considered synergistic combinations of parameters that simultaneously improve the quality of geometry and strength of the structure. Hence, the synergy of build orientation with infill percentage, with clear multiple benefits. Increased infill in on-edge (Y axis) builds not only causes more extruded filaments to line up with the loading direction for improved tensile strength under stress transfer and inter-layer adhesion, but also for better internal thermal warp resistance for dimensional stability, mostly in the Z axis. On the other hand, the fusion of a flat build tends to be lower and the distortion higher when infill is low, resulting in lower strength and accuracy. Similarly, infill percentage in combination with the nozzle diameter affects both properties. The results have indicated that an increase of infill level, coupled with deliberate use of larger nozzle size (0.5 mm), provides a smoother fiber deposition with wider filament strands with good inter-bead overlap, which improves mechanical properties and minimizes geometrical irregularities. But with low infill, nozzle size is of less importance because insufficient internal structure favors void formation and thermal instability, which negatively impacts strength and accuracy.

4.5 Validation through Confirmation Test and Development of Regression Model

Because the initial Taguchi L18 orthogonal array did not include an optimal combination of process parameters, a confirmation test was used to verify the predicted optimal settings. In this experiment, the parameters that were determined to be the most influential ones by ANOVA were 100 % infill percentage (A3), 0.5 mm nozzle diameter (F3), and on-

edge build orientation (D2) used for the multiple linear response functions. The parameters which showed significant contributions in terms of statistically significant values are predicted to be optimal for increasing both the dimensional accuracy and tensile strength.

To assess the statistical reliability of this optimization, the 99% confidence interval (CI) was estimated following the standard formula (Ref 37) given as follows:

$$CI = \sqrt{f_{\alpha} \cdot V_e \cdot \left(\frac{1}{DOF_m + \sum_{i=1}^{n_p} DOF_i} \right)} \quad (\text{Eq 16})$$

where $F_{\alpha} = 10.04$, which is the F -value at 99% confidence for (1, 4) degrees of freedom, $V_e = 0.51$, which is the error variance from ANOVA, $DOF_m = 1$, and $\sum_{i=1}^{n_p} DOF_i = 5$, which includes

the degrees of freedom for the significant factors D, F, and A. Substituting into the formula, the confidence interval was found to be ± 0.92 dB. The predicted S/N ratio at the optimum settings (Y_{opt}) was calculated using the following equation (Ref 37).

$$Y_{\text{opt}} = T/N + (f_{D_{\text{opt}}} - T/N) + (f_{F_{\text{opt}}} - T/N) + (f_{A_{\text{opt}}} - T/N) \quad (\text{Eq 17})$$

where $T/N = 33.67$ dB is the grand mean of the S/N ratios. The optimal S/N ratios of build orientation, nozzle diameter, and infill percentage are $f_{D_{\text{opt}}} = 34.25$, $f_{F_{\text{opt}}} = 33.37$, and $f_{A_{\text{opt}}} = 33.38$, respectively. The resulting predicted S/N ratio was:

$$Y_{\text{opt}} = 33.67 \text{ dB} \quad (\text{Eq 18})$$

Thus, the expected range for the confirmation test lies between 32.74 dB and 34.59 dB, at a 99% confidence level.

Table 8 summarizes the comparison of S/N ratios obtained from three different sources: linear regression modeling, ANOVA-based prediction, and the confirmation experiment. The S/N ratio from the confirmation test was 35.28 dB, which

Table 8 Comparison of S/N ratios

Source	S/N ratio, dB	99% Confidence interval, dB
Linear regression	36.47	...
Predicted by ANOVA	33.67	32.74 to 34.59
Confirmation experiment	35.28	...

Table 10 Regression model summary for tensile strength prediction

Predictor	Coefficient (β)	Standard Error (SE)	p value
Intercept (β_0)	- 28.012	3.21	0.0005
Build orientation (D, β_1)	13.842	2.16	0.0003
Nozzle diameter (F, β_2)	9.617	1.78	0.0006
Infill percentage (A, β_3)	3.478	0.94	0.0020

Table 9 Comparison of dimensional accuracy and tensile strength values obtained from linear regression model and confirmation experiment, including the 99% confidence interval range

Source	Avg. dimensional error, %	Maximum deviation, mm	Tensile strength, MPa	99% Confidence interval, MPa
Linear regression	0.14%	0.06	61.4	...
Confirmation experiment	0.16%	0.07	58.4	54.9 to 62.3

slightly exceeds the upper bound of the computed 99% confidence interval [32.74 dB, 34.59 dB]. This minor deviation may be attributed to inherent experimental variability and the use of mean response values in the prediction model. Despite this, the proximity of the confirmation result to the predicted range reinforces the robustness and reliability of the optimization methodology. Moreover, the linear regression model, which predicted an S/N ratio of 36.47 dB, overestimated the outcome, likely due to its broader sensitivity to coded parameter values. Nonetheless, the proximity of all three values indicates a high degree of agreement and model validity.

Table 11 Microstructural analysis of specimens with optimized parameters

Specimen	Infill percentage, %	Nozzle diameter, mm	Build orientation
S1	100	0.5	Y (on-edge)
S2	80	0.5	Y (on-edge)
S3	100	0.3	X (flat)

Moreover, Table 9 shows the comparison between the measured tensile strengths. The strength from the confirmation experiment was 58.4 MPa, which was close to the 99% confidence range (54.9-62.3 MPa). The predicted results of the regression model indicated 61.4 MPa, which is a little bit more than the actual result, but within the confidence intervals, reinforcing the credibility of the optimization procedure. Dimensional parameters were measured, as well. The confirmation trial demonstrated the average dimensional error of 0.16% and the maximum deviation of only 0.07 mm, confirming the fact that the optimal parameter combination can deliver excellent geometric fidelity in addition to strength. A multiple linear regression model was also developed to quantitatively predict the tensile strength from significant parameters. The model is expressed as follows.

Tensile Strength(MPa) = $\beta_0 + \beta_1 \cdot D + \beta_2 \cdot F + \beta_3 \cdot A$

(Eq 19)

where β_0 is the intercept; $\beta_1, \beta_2, \beta_3$ are the coefficients; D, F, A are the build orientation, nozzle diameter, and infill percentage, respectively. With the estimated coefficients, the multiple linear regression model is expressed as:

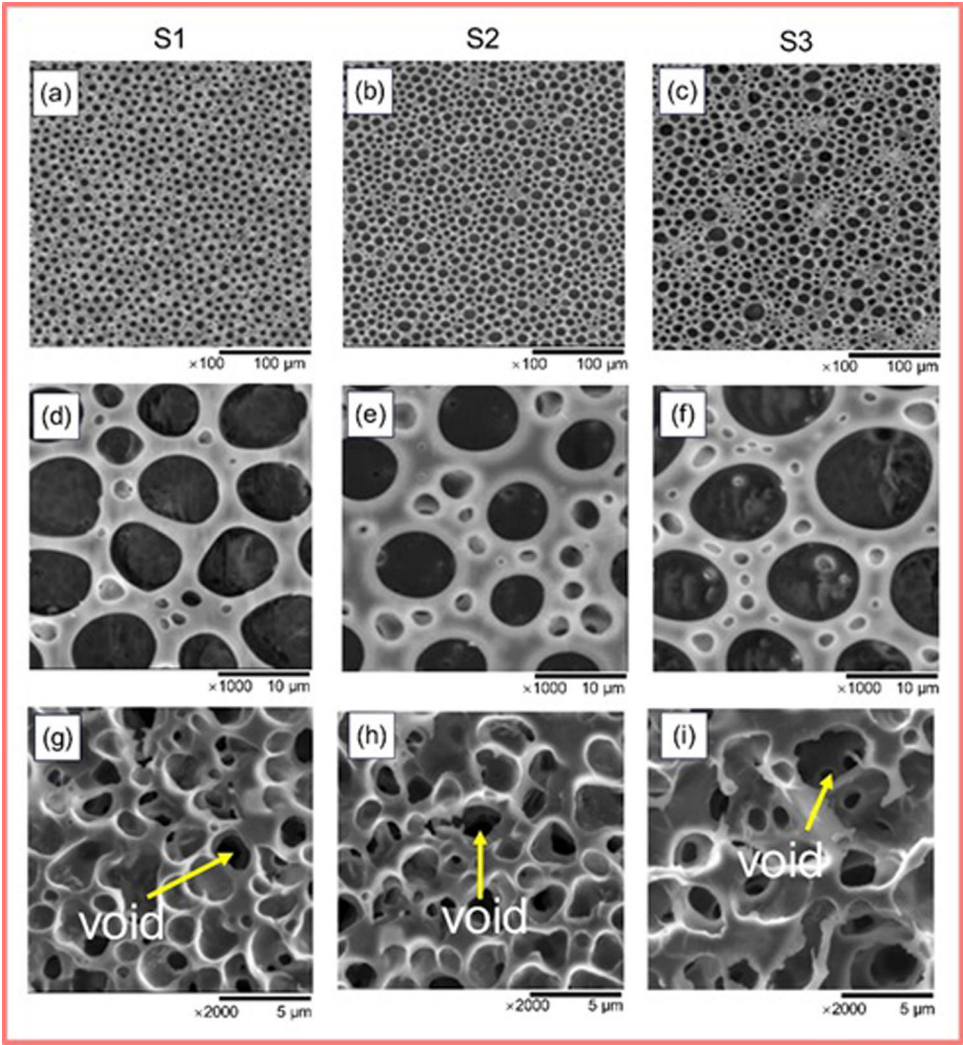


Fig. 12 SEM images of specimens

$$\text{Tensile Strength(MPa)} = -28.012 + 13.842 \cdot D + 9.617 \cdot F + 3.478 \cdot A \quad (\text{Eq } 20)$$

As reported in Table 10, the regression coefficients were statistically significant with p values less than 0.005 and the following standard errors. With $R^2 = 0.88$, the model shows good predictive accuracy. It is a reliable instrument for predicting the tensile performance at various combinations of process parameters within the experimental domain. To ensure validity, the model assumptions were rigorously tested. The normality of residuals was confirmed using the Shapiro–Wilk test (Ref 38) ($p = 0.213$), which indicates no significant deviation from normality.

4.6 Microstructural Analysis

To support the mechanical test results and explore the influence of FDM process parameters on material integrity, micro-structural analysis was carried out on three selected BB-PLA specimens using scanning electron microscopy and transmission electron microscopy. These analyses were based on surface topography, pore content, and inter-layer bonds. Three specimens (S1, S2, and S3) were chosen to represent varying degrees of optimization of the three factors with the most significant effects (infill percentage, nozzle diameter, and build orientation) as identified by ANOVA. The configuration of the specimens is summarized in Table 11. The surface quality and porosity were investigated by scanning electron microscopy through a HitachiTM1000 operating at different magnification levels. Both S1 and S2, which were printed in the on-edge mode, were rather flat, with only a few significant voids that are visible in Fig. 12. The evenness is believed to be caused by infills and larger nozzle sizes, which have allowed even deposition of the materials and better fusion between layers. The part with a smaller nozzle and reduced infill has a mild surface texture, which has little impact on the overall part uniformity.

In contrast, specimen S3 got the smallest nozzle diameter (0.3 mm) and flat orientation and showed the most visible surface imperfections and highest porosity. Scanning electron microscope images showed the formation of discontinuous inter-layer lines and a large amount of micro-voids, presumably caused by inadequate material deposits and a decreased

temperature during the fusion of layers. Finally, the morphology has identified that there are internal inconsistencies resulting in weaker mechanical properties. The SEM micrographs reveal significant variations in void distribution and micro-structural homogeneity for S1, S2, and S3. S2 exhibits the most homogeneous structure with the smallest and less frequent voids, as depicted in Fig. 12(b), (e), and (h), which signifies better inter-layer bonding and lower porosity. This observation is directly correlated to its superior tensile strength (59.84 MPa) and lowest dimensional error (0.16%), and, hence, affirms that improved layer fusion is responsible for both mechanical robustness and geometric fidelity. In contrast, S3 exhibits larger and more extensive voiding (Fig. 12c, f, and i) and occurs in a material that is poorly bonded and has material discontinuity, as demonstrated by the lower tensile strength and higher dimensional error. S1 is intermediate in character, both in terms of void content and corresponding medium level of mechanical property. These results support the conclusion that micro-structural quality, especially void suppression and fusion continuity, is a major determining factor of the performance of a FDM part.

Further, Philips CM 100 transmission electron microscope was used in order to study the inter-layer bond and phase continuity. Figure 13 reveals that both specimens S1 and S2 show well-formed and continuous inter-layer bonds with no noticeable phase separation so that the molecular adhesion is remarkable and the fusion in print is stable. Specimen S3, in contrast, shows interfacial gaps and irregularities along the boundaries indicating fewer thermal bonds and connections between the layers. In addition to that, the TEM images offer further information on the internal morphology and crystallite distribution of S1, S2, and S3. S2 has the most homogeneous and fine-grained internal microstructure (Fig. 13b) and a well-developed and uniformly distributed crystalline phase. This microstructure is related to its ultimate tensile strength (59.84 MPa) and low dimensional error (0.16%) indicating that finer size of crystallite networks improves bonding between layers and imparts rigidity to the material. In contrast, S3 (Fig. 13c) shows aggregated, densely packed structures with poorly delineated boundaries reflecting poor crystallite dispersion, which is in line with poor mechanical and dimensional properties. S1 (Fig. 13a) features moderate dispersion of an intermediate microstructure which agrees with its intermediate performance. These results affirm that crystallite uniformity and dispersion, as captured via TEM, are key contributors to the

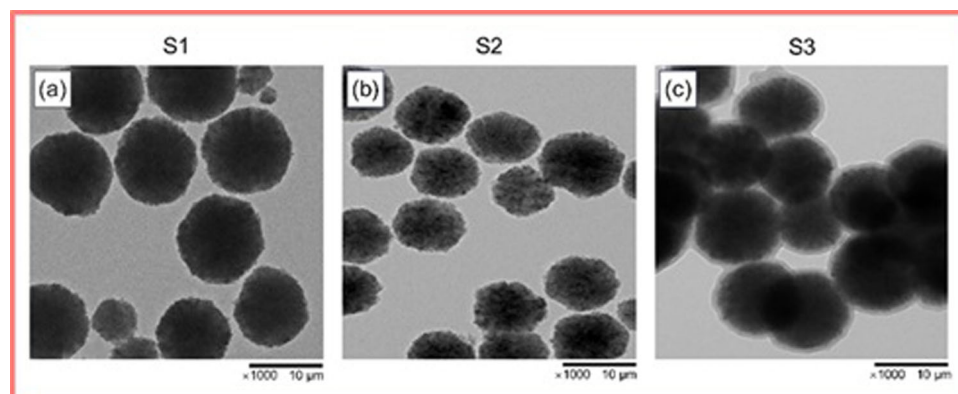


Fig. 13 TEM images of specimens

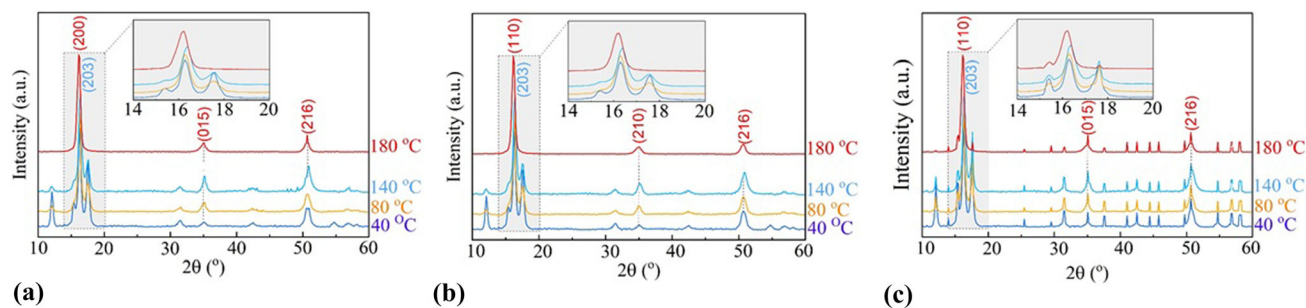


Fig. 14 Comparison of XRD results at various temperatures (a) Specimen S1 (b) Specimen S2 (c) Specimen S3

mechanical and geometric quality of FDM-printed BB-PLA parts.

The microstructural results are in good agreement with the tensile strength trends, validating that, at optimal strength, high infill density, larger nozzle diameters, and on-edge orientation are related. When the parameters go beyond this optimal combination, surface integrity is degraded, interlayer bonding deteriorates, leading to increased porosity and decreased mechanical reliability for BB-PLA printed components.

XRD analysis was performed on BB-PLA specimens that were produced in accordance with different values of FDM process parameters in order to characterize their crystalline structure and phase composition. The objective was to evaluate the effects of changing infill percentages, nozzle diameter, as well as build orientation on crystallinity, which has a direct impact on mechanical and thermal performance. The XRD patterns of the specimens S1, S2, and S3 have been compared in Fig. 14.

Specimen S2, which was printed at 80% infill using a 0.5 mm nozzle in the on-edge orientation, exhibited the sharpest and most intense diffraction peaks (Fig. 14b). The dominant peak at $2\theta = 16.6^\circ$ (assigned to the (110) planes) recorded an intensity of 1484 a.u., accompanied by reflections at 18.8° (203) with 838 a.u., 30.2° (210) at 483 a.u., and 50.1° (216) at 495 a.u. This distinct peak profile is indicative of well-formed crystalline domains, characteristic of the α -form PLA. The enhanced crystallinity is in agreement with the specimen's superior mechanical performance (tensile strength: 59.84 MPa) and minimal dimensional error, confirming the benefits of balanced material deposition and thermal management. Specimen S1, printed at 100% infill with the same nozzle diameter and orientation, showed moderately sharp peaks with slightly reduced intensities (Fig. 14a). The primary peak at 16.5° (200) reached 1078 a.u., while other peaks appeared at 18.6° (203) with 674 a.u., 30.0° (015) with 382 a.u., and 50.2° (216) with 404 a.u. These results suggest a semi-crystalline structure with moderate molecular ordering. Although higher infill promotes material consolidation, the dense deposition may have limited cooling efficiency, reducing the crystallization kinetics. This is reflected in its slightly lower tensile strength (57.92 MPa) compared to S2. Specimen S3, printed with a 0.3 mm nozzle, 100% infill, and flat orientation, displayed broad and weak diffraction patterns (Fig. 14c). The major peak at 16.2° (110) reached only 883 a.u., while the secondary peaks were 18.4° (203) at 516 a.u., 30.1° (015) at 284 a.u., and 50.0° (216) at 288 a.u. The low intensity and broadened peaks reflect limited crystal domain formation, most likely due to rapid cooling, lower material consolidation, and poor thermal retention during deposition. This aligns with the inferior mechanical behavior

(46.25 MPa tensile strength) and higher dimensional error observed in S3.

Overall, XRD analysis proves that the FDM process parameters are important in shaping the crystalline structure of BB-PLA. When printing at a high infill rate with a wide nozzle, on edge, the printed parts are more crystallized, which enhances their utility. It reveals that optimizing parameters is vital to achieve both strong structures and better material structures at the molecular level. The microstructure of BB-PLA components was influenced significantly by the FDM process parameters, as revealed from SEM, TEM, and XRD analyses. The usage of large nozzles, higher filler content, and printing on their side led to better surface finish, interlayer adhesion, and crystallinity. It is having a positive impact on mechanical properties. At non-optimal conditions, a higher porosity and lower interlocking structure were reported. These results confirmed that the physical properties of FDM printed biopolymer specimens such as strength and microstructure can't be improved without improving the process.

4.7 Discussion

The findings of this study offer mechanistic insight into how FDM process parameters govern the simultaneous optimization of dimensional accuracy and tensile strength in BB-PLA components. As it turns out, the performance of printed parts is not solely determined by the material properties, but also from the intricate interplay between thermal gradients, deposition geometry, and interfacial bonding dynamics, all of which are influenced by process settings.

Of the seven studied parameters, build orientation was found to be the most significant factor (42.07%) that contributes to tensile strength. This high dependency is based on the anisotropic nature of FDM printing. The on-edge orientation places the filament layers in the better direction to align more favorably with the direction of tensile load and allows the stress to be passed along the strong polymer chains as opposed to weak interlayers. Moreover, in this orientation, more layers of relatively thin cross sections will be produced for the same overall part height, providing an overall largest interfacial bonding area, which is an important aspect in fracture resistance. This trend aligns with the findings of Hikmat et al. (Ref 33) and Sood et al. (Ref 39) for ABS, although our study highlights a stronger influence in BB-PLA, likely due to its higher crystallization shrinkage (Ref 29), which exacerbates the effects of build orientation on internal stress and bonding.

Nozzle diameter was the second most important factor (24.09%), and it is mechanistically connected to volumetric flow rate and thermal history. A higher nozzle diameter

(0.5 mm) increases the volumetric flow of the molten polymer, which in turn increases the cooling time of the individual deposited bead. This long heat sealing time causes increased mobility of the polymer chains and inter-diffusion between the polymer layers, which further increases interfacial bonding. In addition, XRD measurements showed that samples printed through larger nozzles presented higher degrees of crystallinity (Fig. 14b), confirming how the thermal profile corresponding to higher flow rates favors crystal nucleation and growth, namely, another cause of the enhanced mechanical performance. The relation between microstructure and thermal behavior induced by the nozzle is in agreement with the findings on other semi-crystalline thermoplastics (Ref 35, 40, 41).

Infill percent (18.98%) also had a significant effect mainly by regulating internal density and void structure. Increased infill settings minimize pore volume and increase the load-bearing cross-sectional area, which increases tensile strength and dimensional stability. However, BB-PLA is more sensitive to the effect as it has a higher thermal shrinkage and crystallinity, which could magnify warpage and internal stress if uneven cooling occurs. The dominant influence of infill in this study emphasizes the findings of Johnson and French (Ref 36), highlighting the importance of void reduction in BB-PLA to maintain the progressive development of strength. Importantly, microstructural analysis reinforces these interpretations. The highest tensile strength in T15 correlated with the low porosity found in SEM (Fig. 12e) and the high crystallinity revealed by XRD (Fig. 14b). These results show a synergistic effect of optimal thermal history and geometric orientation conditions that are produced by the specific combination of build orientation, nozzle diameter, and extrusion speed. However, conditions in T3 and T6, which showed high standard deviation and reduced strength, likely caused thermal gradients and poor interlayer fusion, as evidenced by more irregular microstructures.

Although the trends observed for BB-PLA are similar to those for conventional PLA and ABS, this study has shown that bio-based polymers have novel sensitivities due to their compositional and thermal properties. Notably, previous optimization works have usually been focusing on either geometric accuracy or mechanical properties in isolation. Our dual-objective framework illustrates that these outcomes are often interdependent and occasionally competing, highlighting the necessity of co-optimization. Furthermore, the results are consistent with the recent works of Morvayová et al. (Ref 28), who emphasized the need for integrated multi-parameter strategies in biocomposite FDM, and Kaplan et al. (Ref 29), who demonstrated the role of predictive modeling in mitigating parameter-induced variability.

In conclusion, this study provides a mechanistic insight into the influence of process parameters on BB-PLA performance via the interconnected physical phenomena, comprising thermal behavior, anisotropy, void population, and crystallinity. Combining the experimental work with the microstructural validation makes this work advance both the scientific basis and practical applicability of sustainable materials in additive manufacturing.

5. Conclusions

This study aimed to simultaneously enhance dimensional accuracy and tensile strength in FDM of BB-PLA through a comprehensive dual-objective optimization framework. Using a Taguchi L18 orthogonal experimental array, seven critical process parameters were systematically varied, and optimal settings were determined, which yielded a significant improvement in both geometric fidelity and mechanical performance. These outcomes were corroborated by microstructural analyses, revealing enhanced interlayer adhesion and increased crystallinity.

Beyond identifying an optimal process window, the findings advance the field of sustainable additive manufacturing by validating BB-PLA as a viable alternative to petroleum-based polymers for functional, load-bearing components. This study provides a validated framework for using BB-PLA in functional, load-bearing applications, moving it beyond prototyping and contributing to the reduction of petroleum-based plastic consumption.

Notwithstanding the contributions that it has made, the present investigation has discernible limitations. The data were produced using only a Creality Ender 3 printer that used commercially sourced PLA filament under predefined settings in terms of infill density, nozzle diameter, and print orientation. Results may not be generalizable to alternate printer architectures, disparate slicer configurations, or varied ambient conditions. In addition, variations between filament lots, such as moisture content, chromophore additives, and manufacturer-specific formulations can modulate crystallinity and the resulting mechanical properties. As a consequence, people planning to repeat or extrapolate these results must limit their experimental configurations to similar hardware and process parameters, and are encouraged to validate consistency with the help of preliminary calibration prints and systematic quality control evaluations.

Future investigations may focus on the following key aspects:

- Testing the optimized parameters on complex geometries to assess scalability.
- Investigating the interplay between printing parameters and long-term stability (aging, creep) of BB-PLA parts would be valuable.
- Exploring hybrid manufacturing or in-process monitoring techniques (Ref 42) could be a promising avenue to further enhance reliability and control.

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